

---

# Context length and readout parameterisation select the emergence exponent of in-context skills in an additive model

---

Anonymous Authors<sup>1</sup>

## Abstract

We study how a feature-learning model acquires *skills* that are presented in context. Targets are additive models  $y = \sum_p c_p \sigma(\langle v_p, x \rangle)$  with mean-zero task coefficients  $c_p$ , and a student with learned features and a linear-in-label attention readout must recover the directions  $v_p$  by online SGD from prompts of length  $N$ . For a single skill of information exponent  $k^*$  we obtain the population loss in closed form. As  $N \rightarrow \infty$  it is the in-weight loss with the link squared (effective exponent  $2k^*$ , a fact implicit in one-step analyses that we make dynamical). At finite  $N$  the loss acquires an *alignment-dependent* variance term  $\Gamma^2 V(m)/N$ , and the way the readout  $\Gamma$  is parameterised decides what this term does: a readout pinned at order one makes the unaligned state drift-stable below  $m^*$ , so that more prompts cannot substitute for context length; a free scalar readout shrinks to a Wiener factor, which raises the effective exponent from  $2k^*$  towards  $4k^*$  and restores a multiplicative trade-off between context length and training time; a readout tied to the feature norm removes the  $N$ -dependence altogether. A parameter-free population ODE reproduces the exponents measured by online SGD at  $d = 32$  for three readout protocols and four context lengths (11 of 12 cells within 0.6, the censored cell within 1.0). With many skills the dynamics decouple with emergence times  $\propto 1/\pi_p$ ; additive compositions of learned skills follow by linearity. In small softmax transformers we find a  $\approx 3\times$  per-token handicap of short contexts in the small-alignment regime and the opposite ordering when the initial alignment is large; these observations are consistent with, but do not test, the mechanism.

---

<sup>1</sup>Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

## 1. Introduction

Theories of feature learning explain *when* a network acquires a direction  $v$  in input space in terms of the information exponent  $k^*$  of the link function (Ben Arous et al., 2021): online SGD needs  $d^{k^*-1}$  samples ( $d \log d$  for  $k^* = 2$ ). Theories of emergence and scaling laws (Michaud et al., 2023; Nam et al., 2024; Ren et al., 2025) then sum many such acquisitions over a power-law dictionary of skills. Both lines treat the skill as something learned *in the weights* from a fixed task. Large models, however, acquire much of their behaviour as *in-context* skills: the pretraining data presents a skill through prompts whose labels are generated by a task-dependent coefficient, and the network must learn features that make the skill recoverable from a context of length  $N$ . Recent analyses of in-context learning with learned features (Oko et al., 2024; Nishikawa et al., 2025; Kim & Suzuki, 2024; Gu et al., 2026) treat the context length as an *inference-time* resource: it controls how well a trained model estimates the task, and in one-step analyses of pretraining it enters only through concentration, so that more prompts can substitute for longer prompts (Oko et al., 2024, Remark 3).

We ask a different question: *does the context length, together with how the readout is parameterised, change which exponent governs the acquisition of a skill during pretraining?* We answer it in the smallest setting in which the question is sharp—one or several skills with  $k^* = 2$  (even links, where in-weight learning is easy), a student with features on the sphere and a linear-in-label readout, online SGD—and then test the predictions outside the solvable model.

**Contributions.** (1) **Closed form.** The population in-context loss of a single feature is  $L_A(m) = 1 - 2\Gamma g(m)^2 + \Gamma^2 g(1) \left[ \left(1 - \frac{1}{N}\right) g(m)^2 + V(m)/N \right]$ , where  $m$  is the overlap with the skill direction,  $g(m) = \mathbb{E}[\sigma(\langle v, x \rangle) \sigma(\langle w, x \rangle)]$  and  $V(m) = \mathbb{E}[\sigma^2 \sigma^2]$  (Prop. 1). As  $N \rightarrow \infty$  it is the in-weight loss with  $g \mapsto g^2$ : the effective exponent is  $2k^*$ . The  $m^{2k^*-1}$  gradient is inside the one-step analysis of Oko et al. (2024, Lemma 21) (and depends on parameterisation and initialisation, cf. Nishikawa et al. (2025)); we make the statement dynamical and verify it by online SGD, including the  $k^* = 2$  case excluded from Ren et al.

(2025). (2) **The finite-context term and the readout** (Props. 2–3).  $\Gamma^2 V(m)/N$  grows with alignment. With  $\Gamma$  pinned at order one the origin is drift-stable below  $m^*$ ,  $m^{*2} = 16\Gamma/(N(8-4\Gamma(1-\frac{1}{N})-24\Gamma/N))$ ; under SGD the threshold is soft, and more prompts per step do not remove it (Fig. 3). With a free scalar readout,  $\Gamma$  moves towards the Wiener value  $\Gamma^*(m) = g^2/(g(1)[(1-\frac{1}{N})g^2 + V/N])$  and multiplies the feature gradient: the effective exponent rises from  $2k^*$  towards  $4k^*$  as the context statistic becomes noise-dominated; in the adiabatic limit the escape time scales as  $1/N$  there—the multiplicative trade-off of one-step analyses (Oko et al., 2024, Remark 3) reappears as a consequence of shrinkage—and at finite readout learning rates the ODE gives an intermediate, parameter-dependent  $N$ -dependence that we test prospectively. With the readout tied to the feature norm (the 2-homogeneous student of Ren et al. (2025)) the directional dynamics are norm-independent (their Lemma B.1) and the  $N$ -dependence disappears. The population ODE predicts the exponents measured by SGD at  $d = 32$  (Fig. 1). (3) **Many skills and composition.** With a power-law dictionary the dynamics decouple with  $T_p \propto 1/\pi_p$  (measured slope 1.44–1.53 for  $\pi_p \propto p^{-1.5}$ ) against 0.75–0.89 for two in-weight baselines, a corollary of (1); additive compositions of learned skills emerge with the slower skill, by linearity. (4) **Transformers.** In a two-layer softmax transformer with small initial alignment ( $d = 256$ , width 32) a context of 16 with  $16 \times$  the prompts emerges  $\approx 3 \times$  later than a context of 256 at equal tokens per step; with large initial alignment ( $d = 32$ , width 256) the ordering reverses. Neither pre-registered extreme (a trap; interchangeability) held; we report both as observations.

## 2. Setting

Inputs  $x \sim \mathcal{N}(0, I_d)$ . A *skill* is a unit vector  $v_p \in S^{d-1}$  with link  $\sigma$ ; we use the normalised Hermite polynomials  $\sigma_k = \text{He}_k/\sqrt{k!}$ , so  $\mathbb{E}[\sigma_j(\langle u, x \rangle)\sigma_k(\langle w, x \rangle)] = \delta_{jk}\langle u, w \rangle^k$  for unit  $u, w$  and  $k^* = k$ . A *task* is a coefficient vector  $c$  with independent mean-zero entries,  $\mathbb{E}c_p^2 = s_p^2$ ; a *prompt* consists of  $N$  pairs  $(x_i, y_i)$  with  $y = \sum_p c_p \sigma(\langle v_p, x \rangle)$  and a query  $(x_q, y_q)$  from the same task. With a dictionary of  $P$  skills and single-skill prompts, skill  $p$  appears with frequency  $\pi_p$ .

**Model A (in-context).** Features  $\phi_j(x) = \sigma(\langle w_j, x \rangle)$ ,  $w_j \in S^{d-1}$ , and a linear-in-label readout

$$\hat{y}_q = \sum_j \Gamma_j \phi_j(x_q) A_j, \quad A_j = \frac{1}{N} \sum_{i=1}^N y_i \phi_j(x_i), \quad (1)$$

the MLP-plus-linear-attention architecture of Oko et al. (2024). We consider three readout protocols: *fixed*  $\Gamma_j = \gamma$ ; *free*, a trainable scalar; and *tied*,  $\Gamma_j = \|u_j\|^2$  with

$w_j = u_j/\|u_j\|$  (the 2-homogeneous student of Ren et al. (2025)). **Model B (in-weight)** is ordinary regression  $\hat{y} = \sum_j a_j \phi_j(x)$  on the fixed task  $c \equiv a$ . Both are trained by online spherical SGD on the square loss with fresh data at every step:  $B$  prompts per step, step size  $\eta$  on the features (renormalised to the sphere after each step), step size  $\eta_\Gamma$  on a free readout initialised at  $\Gamma_0$ , and for the tied readout an unnormalised  $u$  with  $\|u_0\|^2 = \rho_0$ . We report escape times  $T_{0.5}$  (steps until  $|m| \geq \frac{1}{2}$ ) and their *flow-time* version  $T_{0.5} \eta$ , which is what the population ODEs predict; the single-feature experiments initialise at  $m_0 = c d^{-1/2}$  exactly.

Write  $g(m) = \sum_k \alpha_k^2 m^k$  for  $\sigma = \sum_k \alpha_k \sigma_k$  and  $V(m) = \mathbb{E}[\sigma(\langle v, x \rangle)^2 \sigma(\langle w, x \rangle)^2] = \sum_j \beta_j^2 m^j$  where  $\sigma^2 = \sum_j \beta_j \sigma_j$ . For  $\sigma_2$ :  $g = m^2$ ,  $V = 1 + 8m^2 + 6m^4$ .

## 3. A single skill

**Proposition 1** (Population loss). *For  $P = 1$ , one feature at overlap  $m = \langle w, v \rangle$ , readout  $\Gamma$  and  $\mathbb{E}c^2 = s^2$ ,*

$$\begin{aligned} L_A(m) &= s^2 \left[ 1 - 2\Gamma g(m)^2 + \Gamma^2 g(1) \left( \left(1 - \frac{1}{N}\right) g(m)^2 + \frac{V(m)}{N} \right) \right], \\ L_B(m) &= 1 - 2a g(m) + a^2 g(1). \end{aligned} \quad (2)$$

*Proof sketch.* Conditional on the task,  $A = c g(m) + \xi/\sqrt{N}$  with  $\mathbb{E}\xi = 0$ ,  $\mathbb{E}\xi^2 = c^2(V - g^2)$ ;  $x_q$  is independent of the context, and  $\mathbb{E}[\sigma(\langle v, x_q \rangle)\sigma(\langle w, x_q \rangle)] = g(m)$ ,  $\mathbb{E}[\sigma(\langle w, x_q \rangle)^2] = g(1)$ . Expanding  $\mathbb{E}(y_q - \hat{y}_q)^2$  gives (2).  $\square$

We verified (2) by comparing the Monte-Carlo drift  $-\mathbb{E}[\partial_w L] \cdot v$  with  $-\partial_m L_A$  at  $m \in \{0.05, 0.1, 0.2, 0.4\}$  for  $k^* \in \{1, 2, 3\}$ : all 24 points agree within 2.3 s.e. and within 3% for  $m \geq 0.2$  (Appendix).

**The link is squared.** As  $N \rightarrow \infty$ ,  $L_A = s^2[1 - 2\Gamma g^2 + \Gamma^2 g(1)g^2]$  depends on  $m$  only through  $g(m)^2$ , whereas  $L_B$  depends on it through  $g(m)$ . Near  $m = 0$  the drift is  $2\Gamma s^2(2 - \Gamma g(1)) k^* \alpha_{k^*}^4 m^{2k^*-1}$  against  $2ak^* \alpha_{k^*}^2 m^{k^*-1}$  for Model B: the label enters the prediction twice—through the context statistic and through the comparison with  $y_q$ —and the mean-zero task prior removes the cross terms that would provide a degree- $k^*$  path. (A bias added to  $A_j$  would give an in-weight path for the *mean* task only, cf. Zhang et al. (2024) for the linear case; we do not analyse it here.) With the escape heuristics of Ben Arous et al. (2021) (drift  $\eta C m^{\kappa-1}$ , renormalisation drag  $-\eta^2 dm/2$ ) the timescales are  $d^{\kappa-1}$  with  $\kappa = 2k^*$ : for  $k^* = 2$ , in-weight needs  $d \log d$  samples and in-context  $d^3$ .

**Proposition 2** (Finite context: a drift-stable unaligned state). *For  $\sigma_2$  and fixed readout  $\Gamma = \gamma$ ,  $-\partial_m L_A = s^2[(8\gamma -$*

$4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2m/N]$ . For  $\gamma < \gamma_1 = 2N/(N+9)$  the origin is a local minimum of  $L_A$ , the drift is repulsive for  $m < m^*$  and attractive on  $(m^*, 1)$ , with  $m^{*2} = 4\gamma/(2N - (N+5)\gamma) \approx 2\gamma/(N(1-\gamma/2))$  (for  $\gamma \geq \gamma_1$  the overlap decays from every  $m_0 < 1$ ); for general  $\sigma_k$ ,  $k \geq 2$ , the same holds with  $m^{*2(k-1)} \approx 2k\gamma/((2-\gamma)N)$  to leading order, because  $\sigma_k^2$  has normalised second Hermite coefficient  $\beta_2^2 = 2k^2$  (Appendix A).

The variance  $V(m)$  of the context statistic grows with alignment—an aligned feature has heavy tails that coincide with the teacher’s—so the finite-context noise penalises alignment. A feature initialised at  $m_0 \approx d^{-1/2}$  is attracted only if  $d \lesssim d^* = N(8 - 4\gamma(1 - \frac{1}{N}) - 24\gamma/N)/(16\gamma)$  ( $\approx 31$  for  $\gamma = 1$ ,  $N = 128$ ). For  $k^* = 1$  the repulsion is a constant fraction  $\gamma/N$  of the attraction and is harmless. One-step analyses of pretraining do not see this term because the readout is initialised at  $\gamma \propto d^{-k^*}$  (Oko et al., 2024, Lemma 18), which makes  $\Gamma^2V/N$  negligible; the phenomenon lives where  $\Gamma$  is order one or trained.

**Remark 1** (Soft under SGD). SGD is a diffusion on the sphere; a run started below  $m^*$  escapes with a Kramers-type probability  $\sim e^{-\Delta L/D_{\text{eff}}}$ ,  $\Delta L = L(m^*) - L(m_0)$ ,  $D_{\text{eff}} = \eta \text{Var}(\partial_m \text{-gradient})/2$ . At  $\gamma = 1$ ,  $N = 128$ :  $d = 16$  has  $m_0 > m^*$  (no barrier),  $d = 32$  sits on the ridge ( $\Delta L/D_{\text{eff}} \approx 0.02$ ),  $d = 64$  is trapped ( $\Delta L/D_{\text{eff}} \approx 20$ ). Observed escapes in  $3 \cdot 10^5$  steps: 3/3, 1/3, 0/3. Larger batches lower  $D_{\text{eff}}$  and deepen the trap.

**Proposition 3** (Readout parameterisation: three regimes—informal summary of Theorems A–B and Prop. 4, Appendix B). Consider the population gradient flow of (2) for  $\sigma = \sigma_2$  in flow time  $t = \eta \cdot$  steps, started at  $m_0 = d^{-1/2}$ , and let  $\tau_{\text{esc}}$  be the time to reach  $m = \frac{1}{2}$ .

(a) Pinned readout  $\Gamma = \gamma$ . The overlap drift is  $(8\gamma - 4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2m/N$  (Prop. 2). If  $m_0 < m^*(\gamma, N)$  then  $\tau_{\text{esc}} = \infty$ ; otherwise  $\tau_{\text{esc}} = \Theta(m_0^{-2})$  with an  $N$ -dependence that saturates for  $N \gg \gamma d$ . The condition involves  $N$  and  $\gamma$  only—not the number of prompts per step, which does not enter the population drift.

(b) Free scalar readout  $\dot{\Gamma} = -r \partial_{\Gamma} L_A$ ,  $r = \eta_{\Gamma}/\eta$ ,  $\Gamma(0) = \Gamma_0$ . The fixed point of  $\Gamma$  at given  $m$  is  $\Gamma^*(m) = g^2/(g(1)[(1 - \frac{1}{N})g^2 + V/N])$ , approached at rate  $2r g(1)[(1 - \frac{1}{N})g^2 + V/N] \approx 2r/N$ . Adiabatic case ( $2r/N \gg$  the overlap rate): the overlap sees  $L_A(m, \Gamma^*)$ , whose exponent is  $4k^*$  for  $m^{2k^*} \ll V/N$  and  $2k^*$  otherwise, and in the former regime  $\tau_{\text{esc}} \propto m_0^{-(4k^*-2)}/N$  (Appendix A). Growth-limited case ( $2r/N \lesssim$  the overlap rate; E2 at  $N = 16$  is adiabatic to 1%, the other cells are not):  $\Gamma$  lags  $\Gamma^*$  and  $\tau_{\text{esc}}$  depends on  $(N, \Gamma_0, r)$  jointly; the ODE gives intermediate  $N$ -dependences (e.g.  $\tau(16)/\tau(256) = 4.4$  at

$d = 64$ ,  $\Gamma_0 = 0.01$ ,  $r = 1$ , against 16 in the adiabatic  $1/N$  law).

(c) Tied readout  $\Gamma = \rho = \|u\|^2$ , features  $\sigma(\langle u/\|u\|, x \rangle)$ . The directional velocity is  $(1 - m^2)[4gg' - \rho(2(1 - \frac{1}{N})gg' + V'/N)]$ , which is exactly the pinned velocity at  $\gamma = \rho$  divided by  $\rho$  (norm-independence of the attraction, Ren et al., 2025, Lemma B.1): a tied readout behaves like a pinned readout of value  $\rho(t)$  under a time change. From a small  $\rho_0$ ,  $\dot{\rho} = -4\rho \partial_{\rho} L_A$  leaves  $\rho$  near  $\rho_0$  and  $\tau_{\text{esc}} = \Theta(m_0^{-2})$  is independent of  $N$  provided  $\rho_0 \ll \rho_{\text{trap}}(m_0, N)$  (Theorem A(iii);  $\rho_{\text{trap}} \propto m_0^{2k^*-2}$ , so this window narrows quickly with  $k^*$ , E6)—within it the tied readout self-selects the small-readout regime. Let  $\rho^*(m) = 2Nm^2/(4 + (N+5)m^2) \approx Nm^2/2$  be the value at which the directional drift vanishes. For  $\rho_0 > c_2\rho^*(m_0)$ ,  $c_2 = \frac{4}{3}$  (Theorem A(iii); between  $\rho^*$  and  $c_2\rho^*$  the overlap first falls and then escapes), the population flow traps permanently ( $m \propto t^{-2}$  while  $\rho \propto t^{-1}$ , so  $\rho$  never falls below the moving threshold); under SGD, noise keeps  $m$  at the noise floor while  $\rho$  decays, and the feature is eventually released—a long delay rather than a trap (E5).

Integrating the population flow confirms the three regimes quantitatively (Appendix A): for  $\sigma_2$  at  $d = 512$  and  $N = 32, 128, 512$  the escape times are  $\infty, 771, 386$  (pinned  $\gamma = 0.1$ ),  $9.0 \cdot 10^4, 2.2 \cdot 10^4, 5.4 \cdot 10^3$  with  $\tau N = 2.9, 2.8, 2.75 \times 10^6$  (free), and 39, 34, 33 (tied); at  $d = 32$  the statistic is signal-dominated for  $N \gtrsim d^{k^*} = 1024$  and  $\tau = 52, 37, 34, 33$  for  $N = 32\text{--}2048$ . The static value  $\Gamma^*$  is the ridge/Wiener optimum that appears in replica analyses of trained models (Gu et al., 2026, eqs. (297)–(298)); our point is its dynamical effect: the same loss yields a trap, a multiplicative  $N$ – $T$  trade-off (that of Oko et al. (2024, Remark 3)) with an exponent crossover, or no  $N$ -dependence, according to the readout. That a fast readout doubles the exponent of an in-weight problem is known (Bietti et al., 2023, §5); what is specific to the in-context problem is that the finite-context noise drives the readout towards  $\Gamma^*$ , so that whether context length trades off against training time, is irrelevant, or is a hard requirement is decided by how the readout is parameterised. One-step analyses have a readout initialised at  $\gamma \propto d^{-k^*}$  (Oko et al., 2024, Lemma 18), which makes  $\Gamma^2V/N$  negligible and places them in the regime where only the product of prompts and prompt length matters.

**E1: exponents by SGD.** Online SGD,  $\sigma_2$ ,  $d = 32$ ,  $\eta = 1/d^2$ , initial overlap  $m_0 = c d^{-1/2}$ ,  $c \in \{0.5, 0.7, 1, 1.4\}$ , three seeds each; the escape time from  $m_0$  scales as  $m_0^{-(\kappa-2)}$  for  $\kappa > 2$ , so the slope of  $\log T_{0.5}$  against  $\log m_0$  estimates  $\kappa_{\text{eff}} = 2 - \text{slope}$  (12 runs per cell; the estimator is coarse and the censored cells are biased low). Fig. 1 shows  $\kappa_{\text{eff}}$  for free ( $\eta_{\Gamma} = \eta$  and  $10\eta$ ) and tied readouts at

$N \in \{32, 128, 512, 4096\}$  against the population ODE of Props. 1–3 (integrated with LSODA,  $\text{rtol } 10^{-10}$ ; an earlier Euler integrator overestimated times by 0.6–3.6% without changing any exponent or trap verdict) evaluated *after the run* at the same  $d$ ,  $m_0$ -grid,  $\Gamma_0$  and  $\eta_\Gamma/\eta$  (the ODE itself was fixed before the run; its evaluation at these parameters was not pre-registered): tied 4.10–4.44 (ODE 4.15–4.18); free 4.8–6.6 rising as  $N$  falls (ODE 4.85–7.0). With a censored (Tobit) fit for the three cells that contain censored runs, all twelve cells agree with the ODE within 0.5 (ordinary least squares on reached runs only: eleven within 0.6, the free  $N = 32$ ,  $\eta_\Gamma = \eta$  cell 1.0 low); the estimator is a secant over a factor 2.8 in  $m_0$  and local slopes vary, so these are effective, not asymptotic, exponents, and the prompts per step were 32 for  $N \leq 128$  and 8 for  $N \geq 512$  (a  $B = 8$  control at  $N = 128$  gives  $5.38 \pm 0.26$  vs  $5.59 \pm 0.20$ ). The ODE also reproduces absolute escape times: over the 45 fully-reached E1 cells the ratio median-SGD/ODE lies in  $[0.64, 1.34]$  with 24 within 10% and 35 within 20% (no trend in  $N$  or protocol; the largest deviations are at the smallest  $m_0$ ), and one censored cell (free,  $10\eta$ ,  $N = 128$ , smallest  $m_0$ ) is at least  $2.3\times$  slower than predicted (Appendix C). The in-weight model at the same  $k^*$  gives a logarithmic  $m_0$ -dependence ( $\kappa = 2$ ).  $\kappa_{\text{eff}}$  is a secant over a factor 2.8 in  $m_0$  at fixed  $(d, N, \Gamma_0, \eta_\Gamma/\eta)$ , not an asymptotic exponent: at fixed  $N$  and  $d \rightarrow \infty$  the free-readout flow gives  $4k^*$  for every  $N$  with  $N$  only in the prefactor (Theorem B:  $\tau_{\text{esc}} \sim e^{(4k^*-2)k^*2\Gamma_0^2/r} \mu_0^{-(4k^*-2)} d^{2k^*-1} / (4k^*(4k^*-2)N)$ ), while a  $d$ -free limit exists under the joint scaling  $\lambda = N/d^{k^*}$ ,  $\tilde{r} = (\eta_\Gamma/\eta)/d$ , giving  $\kappa(\lambda, \tilde{r}, \Gamma_0)$  whose  $d \rightarrow \infty$  values (6.45/5.40/4.95/4.83) are within 0.06 of the  $d = 32$  ODE numbers; its large- $\lambda$  end at small  $\Gamma_0$  is  $2k^* + 1$  (readout growth from  $\Gamma_0$ ), not  $2k^*$ . The pre-registered “4  $\rightarrow$  8” was this corner structure misstated as  $d \rightarrow \infty$  limits. Two pre-registered predictions in this experiment failed and are reported as such: the learned  $\Gamma$  at  $T_{0.5}$  is 0.18–0.25, below  $\Gamma^*(0.5; N) = 0.38$ –0.99 (the readout has not equilibrated at  $\eta_\Gamma = \eta$ ; the exponent prediction does not depend on this), and a scan of  $T_{0.5}$  against  $d \leq 48$  at  $N = 128$  versus 2048 showed no  $N$ -dependence (slopes  $3.49 \pm 0.08$  vs  $3.70 \pm 0.21$ ), which is what the ODE predicts for  $d \leq 48$  but means the  $N$ -dependence is established only through  $\kappa_{\text{eff}}(N)$  at fixed  $d$ . With the tied readout at  $\rho_0 = 1$ ,  $N = 128$ , the natural O(1)-readout version of Prop. 2,  $d = 32$  escaped (3/3) while  $d = 64$  was trapped (0/3,  $\rho$  decaying to 0.18).

**E2: the three regimes by SGD (the prospective test of Prop. 3).** To test Prop. 3 directly (Fig. 2) we fix  $d = 64$ ,  $m_0 = d^{-1/2}$ ,  $\eta = 1/d^2$ , vary  $N \in \{16, 64, 256\}$  under two schemes—tokens per step fixed ( $NB = 4096$ ) and prompts per step fixed ( $B = 64$ )—and run the four protocols, three seeds each, with the population escape times computed before the run. Pinned  $\gamma = 1$ : no es-

cape at  $N \leq 64$  in  $10^6$  steps (0/9, including  $B = 256$ ), 3/6 at  $N = 256$  ( $d^* \approx 63$ ); pinned  $\gamma = 0.1$ : median 396k/192k/179k steps against 400k/193k/175k predicted; free: 1.78M/614k/402k against 1.84M/627k/414k, ratio  $T(16)/T(256) = 4.4$  (3.6–5.0 over seed pairings) as predicted (4.44)—the growth-limited value of Prop. 3(b), not the adiabatic  $1/N$  law; tied: 17.0k/15.1k/14.0k against 17.0k/16.4k/16.3k (flat within  $\times 1.25$ , seed pairings 0.9–1.45). Three seeds resolve medians to roughly 5%. In every protocol the two batch schemes give the same step counts, as they must at fixed step size (the population drift does not contain  $B$ ); the trade-off is between context length and the number of SGD steps, and a step size scaled with  $B$  is tested in Appendix C. (The free-readout  $N = 16$  cells exceeded the pre-registered  $10^6$ -step cap, as the ODE had predicted, and were rerun to  $2.5 \cdot 10^6$  steps.)

**E5: controls.** Varying the readout’s initial value and learning rate (free:  $\Gamma_0 = 0.1$  or  $\eta_\Gamma = 10\eta$  at  $d = 64$ ) changes the  $N$ -ratio from 4.4 to 9.2 and 10.7 (ODE 9.1 and 8.9; medians within 2% except the  $\eta_\Gamma = 10\eta$ ,  $N = 16$  cell, 24% slow), confirming that the free-readout exponent is a function of  $(N, \Gamma_0, \eta_\Gamma/\eta)$ , as Prop. 3(b) states. A pinned  $\gamma = 0.01$  readout is as  $N$ -flat as the tied one (ratio 1.06, ODE 1.04). A tied readout started at  $\rho_0 = 0.3 > \rho_{\text{trap}}(m_0) = 0.18$  at  $N = 16$  was pre-registered to trap (the deterministic flow provably does, Theorem A(iii):  $m \propto t^{-2}$ ,  $\rho \propto t^{-1}$ ); under SGD all three seeds escaped after a  $\approx 30\times$  delay, once  $\rho$  had decayed to 0.012—so under SGD tying converts the pinned trap into a delay, and the pre-registered wording “the repulsion survives tying” is withdrawn in favour of “is delayed by tying”. Finally, scaling the step size with the batch ( $\eta \propto B$ , pinned  $\gamma = 0.1$ ,  $N = 64$ ) leaves the escape flow time invariant to 0.5% while steps scale as  $1/\eta$ : the irrelevance of prompts per step in E2 is a statement about flow time, independent of how  $\eta$  is chosen.

**E6: a  $k^* = 3$  check (Exp. 9, pre-registered P22).** With  $\sigma = \sigma_3$  ( $g = m^3$ ,  $V = 1 + 18m^2 + 54m^4 + 20m^6$ ),  $\eta = 2 \cdot 10^{-4}$ ,  $B = 64$ ,  $d \in \{8, 16, 32\}$ , three seeds: the tied readout ( $\rho_0 = 0.01$ ,  $N = 128$ ) gives secant exponents 5.7 and 6.0 (ODE 6.5, 6.4)—the  $2k^*$  rule at  $k^* = 3$ , against  $\approx 4$  at  $k^* = 2$ —and medians within 15% of the flow once the two declared post-hoc controls are applied (the  $d = 8$  excess of 18% vanishes at  $\eta/4$ , ratio 1.007; the  $d = 32$  deficit of 20% at  $B = 64$  becomes an excess of 14% at  $B = 256$ , i.e. noise-assisted escape). The pinned readout  $\gamma = 0.2$  has  $m^* = 0.30$ :  $d = 8$  ( $m_0 = 0.35$ ) escapes at  $1.02\times$  the flow time and  $d = 16, 32$  are censored 3/3 at  $10^6$  steps, as predicted. The pre-registered  $N$ -flatness of the tied readout *failed* at  $N = 16$ ,  $d = 16$ : the flow predicts  $1.6\times$  the  $N = 128$  time, the seeds gave 47k, 500k and censored, and at  $d = 32$  all three seeds trapped (predicted:  $\rho_0 = 0.01 > \rho_{\text{trap}} = 0.006$ ). With  $k^* = 3$  the drift

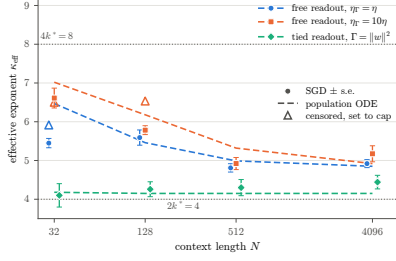


Figure 1. Effective exponent of skill acquisition for  $k^* = 2$  at  $d = 32$ : online SGD (points,  $\pm$ s.e.; triangles: censored runs set to the step cap) against the parameter-free population ODE (dashed). The tied readout stays at  $2k^* = 4$ ; the free readout rises towards  $4k^* = 8$  as the context shortens.

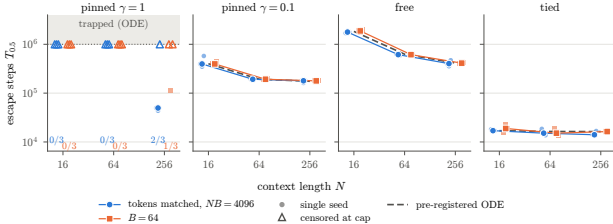


Figure 2. Escape steps  $T_{0.5}$  (first step with  $|m| \geq 0.5$ ) against context length  $N$  for the four readout protocols ( $d = 64$ ,  $\eta = 1/d^2$ ,  $m_0 = d^{-1/2}$ ; shared log-log axes). Light markers are single seeds (three per cell), solid markers and lines the median over seeds, for tokens per step fixed ( $NB = 4096$ , circles) and prompts per step fixed ( $B = 64$ , squares). Open triangles are runs that had not escaped at the step cap ( $10^6$ ) and are drawn at the cap; the counts under the pinned- $\gamma = 1$  panel are runs escaped out of 3. Dashed lines are the population-ODE escape times (flow time  $\times d^2$ ), fixed before the run; pinned  $\gamma = 1$  has no finite ODE escape time (trapped, shaded beyond the cap). The free-readout  $N = 16$  cells use the reruns to  $2.5 \cdot 10^6$  steps (the pre-registered cap was exceeded by the predicted 1.84M steps). At  $N = 64$  both schemes have  $B = 64$  and are the same three runs.

near  $m_0$  is  $\propto m^5$  and  $\rho_0$  was only  $2.6\times$  below  $\rho_{\text{trap}}$ , so SGD noise decides between escape and a long delay. The statement that survives is the conditional one in Prop. 3(c):  $N$ -flat for  $\rho_0 \ll \rho_{\text{trap}}(m_0, N)$ .

#### 4. Many skills and composition

Fig. 4 summarises Exps. 2–3. With  $P$  orthogonal skills and  $M \geq P$  features the loss splits, when each feature is assigned to one skill, into  $\sum_p \pi_p s_p^2 \ell(m_p)$  with  $\ell$  the single-skill loss; each overlap then follows the single-feature dynamics with rate  $\propto \pi_p s_p^2 \Gamma$ , so  $T_p \propto 1/\pi_p$ . In-weight learning of the same dictionary carries the coefficient  $a_p$  (with  $a_p^2 \propto \pi_p$ ), hence  $T_p \propto 1/a_p \propto \pi_p^{-1/2}$ : in-context learning orders skills by frequency about twice as steeply—a corollary of the squared link, since the in-context drift carries  $\mathbb{E} c_p^2 \pi_p$  and the in-weight drift carries  $a_p$ . **E3** ( $d = 32$ ,  $P = 16$ ,  $\pi_p \propto p^{-1.5}$ ,  $M = 64$ , fixed  $\Gamma = 0.1$ , three seeds)

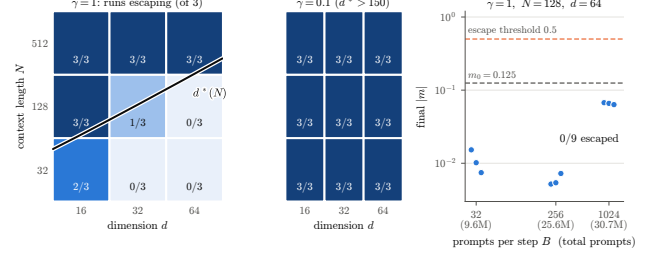


Figure 3. Left: fraction of runs escaping  $m_0 = d^{-1/2}$  within  $3 \cdot 10^5$  steps (fixed readout, 3 seeds per cell; curve:  $d^*(N)$  of Prop. 2; for  $\gamma = 0.1$  the curve lies above the grid). Right: at  $\gamma = 1$ ,  $N = 128$ ,  $d = 64$  no run escapes for  $B \in \{32, 256, 1024\}$  prompts per step (total prompts  $\times 1$ ,  $\times 2.7$ ,  $\times 3.2$ ) and the final overlap falls below  $m_0$ : more prompts do not substitute for context length.

gives slopes of  $\log T_p$  versus  $\log p$  of  $1.44 \pm 0.08$  (drop midpoints;  $1.53 \pm 0.22$  with the 0.5-crossing) for the in-context model, against  $0.75 \pm 0.08$  for a capacity-matched in-weight baseline and  $0.75$ – $0.89$  for the 2-homogeneous in-weight student of Ren et al. (2025) (small initialisation, no hand-chosen readout; the lower value is limited by the logging resolution, the higher one uses a dense early log). Both in-weight baselines fit the target collectively at  $M = 64 > d$  (every skill shared by  $\geq 2$  features, pooled alignment  $\rightarrow 1$ , no single feature above 0.9), so the in-weight slope is an empirical baseline rather than a decoupled-neuron prediction; its emergence times nonetheless obey  $T_p \eta a_p = \text{const}$  (0.23–0.26 across skills and three learning rates), i.e. ordering by the coefficient  $a_p \propto \sqrt{\pi_p}$ , whereas the in-context times obey  $T_p \eta \pi_p m_0^2 = \text{const}$ . The constant  $T_p \eta \pi_p m_0^2 = 0.73$  (IQR 0.70–0.77) across skills and seeds equals the single-feature value (0.64), i.e. the many-skill dynamics is the single-feature dynamics with rate  $\pi_p$ . Per-skill drops are  $\approx 4\times$  sharper than in-weight (relative width 1.08 vs 4.09). The frequency-weighted alignment loss decays as  $t^{-0.40 \pm 0.04}$  over the window in which skills 3–8 emerge, against  $(\alpha - 1)/\alpha = 0.33$  for an idealised staircase; the window spans only a factor 3–8 in time, so the exponent depends on the emergence definition and we do not claim a scaling-law ratio.

**E4: composition.** Because (1) is linear in the context labels,  $\hat{y}_q(c_1, c_2) = \hat{y}_q(c_1, 0) + \hat{y}_q(0, c_2)$  identically, and for independent mean-zero  $c_1, c_2$  the loss on an additive pair is the sum of the single-skill losses. Trained on single-skill prompts only ( $d = 32$ ,  $P = 2$ ,  $\pi = 0.75/0.25$ ,  $M = 4$ ), the pair loss reaches 90% of its drop before or together with the slower skill in every run (never after)—which is forced by linearity, so this is a check of the identity under SGD, not a test; on a thresholded accuracy, where Arora & Goyal (2023) predict an extra delay for tuples, the thresholded pair accuracy is approximately the product of the single-skill accuracies ( $\text{acc}_{12} \approx 0.85 \text{acc}_1 \text{acc}_2 + 0.09$ , r.m.s. 0.008 from

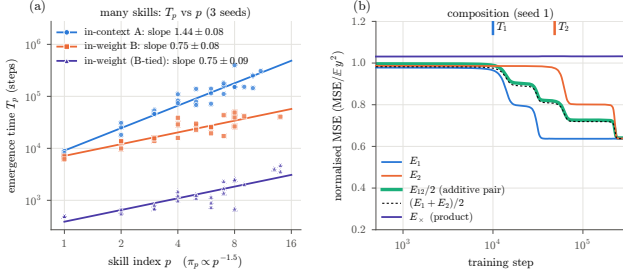


Figure 4. (a) Emergence time (drop midpoint) of skill  $p$  versus skill index for the in-context model (slope  $1.44 \pm 0.08$ ), the capacity-matched in-weight baseline (slope  $0.75 \pm 0.08$ ) and the 2-homogeneous in-weight student (slope  $0.75\text{--}0.89$ ),  $\pi_p \propto p^{-1.5}$ , three seeds each; unlearned skills omitted. (b) Composition (Exp. 3, one seed): normalised MSE on skill-1 prompts, skill-2 prompts, the additive pair (halved) and the product pair; the dotted curve is half the sum of the single-skill losses and lies on the pair curve. Plateaus at 0.64 are the fixed-readout floor  $(1 - \Gamma n)^2$ , not zero.

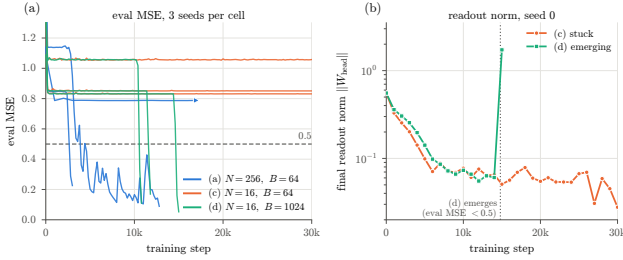


Figure 5. Two-layer softmax transformer,  $d = 256$ , MLP width 32,  $k^* = 2$ . (a) Evaluation MSE for  $N=256, B=64$ ;  $N=16, B=64$ ; and  $N=16, B=1024$  (same tokens per step as the first), three seeds each; one  $N=256$  seed was censored at 17k steps on the plateau. (b) Norm of the final readout for one stuck run ( $N=16, B=64$ ) and one emerging run ( $N=16, B=1024$ , same seed): the norm collapses while the skill is unlearned; the regrowth at emergence is a single logged point (norms logged every 1000 steps).

$y = x$ )—the multiplicative form observed empirically by Okawa et al. (2023), and a delay of the kind Arora & Goyal (2023) describe—while a Gaussian-error closed form fails (heavy tails). A product composition  $\sigma_1(\langle v_1, x \rangle) \sigma_1(\langle v_2, x \rangle)$  is not learned (loss  $\approx 1$ ), as expected for a model never shown product prompts.

## 5. Transformers: where the effect shows and where it does not

We train two-layer softmax transformers (4 heads, pre-LN, Adam, online data; Fig. 5) on the same in-context task with  $k^* = 2$  and ask whether a short context can be compensated by more prompts per step. **Large initial alignment** ( $d = 32$ , MLP width 256; the best of 256 random first-layer directions already has  $|\cos| = 0.43\text{--}0.55$  with  $v$ ): at a fixed 8192 tokens per step the emergence step increases with  $N$

(1000/1400/2800 for  $N = 16/64/256$ )—fewer prompts per step dominate and the mechanism is invisible, as it must be when  $m_0 \gg m^*$ . **Small initial alignment** ( $d = 256$ , width 32;  $|\cos|_0 = 0.13\text{--}0.18$ ):  $N = 16$  with  $B = 64$  prompts never emerges in 30k steps (3/3);  $N = 16$  with  $B = 1024$  (the same tokens per step as  $N = 256, B = 64$ ) emerges in all seeds but at 10.6–14.8k steps,  $\approx 3\times$  later than  $N = 256$  (2.8–4.0k); while a run is on the plateau its final readout norm collapses from 0.57 to 0.03–0.16 (in stuck and emerging runs alike), and the first norm logged after emergence is back at 1.7. Neither pre-registered extreme (a trap, or interchangeability of  $N$  and  $B$ ) holds; the  $\approx 3\times$  ratio rests on two paired seeds, the Adam step was not rescaled with  $B$ , and the readout collapse is identical in stuck and later-emerging runs (Fig. 5b), so it is plateau behaviour rather than a signature of Prop. 3. We state the result as an observation consistent with—not a test of—the mechanism of Sec. 3;  $k^* = 1$  and in-weight controls emerge within 1.4k and 0.4k steps.

**The readout is not the transformer’s knob (Exp. 8, pre-registered P21).** If Prop. 3 transferred through the final linear readout, freezing that readout at its  $O(1)$  initial norm (pinned) should turn the  $N = 16$  handicap into a trap that  $B = 1024$  cannot lift, and freezing it at  $0.1\times$  (pinned small) should rescue  $N = 16, B = 64$ . Neither happened (two seeds per cell, same seeds and code as above): the  $O(1)$ -frozen readout emerged at 5.8k and 8.4k steps with  $B = 1024$ —no later than the trainable one—and the small frozen readout stayed on the plateau for 30k steps at  $B = 64$  in both seeds, while  $N = 256$  emerged under every protocol. The pre-registered kill criterion was met, so we state the negative result: in this architecture the final readout multiplies the residual stream rather than the context statistic, and whatever plays the role of  $\Gamma$  sits inside the attention value/output path, which these protocols did not control. The three-regime picture is therefore established for the solvable model only; Sec. 5 reports the transformer’s  $N$ – $B$  behaviour as an observation.

## 6. Related work

Oko et al. (2024) analyse the architecture (1) with one gradient step; the product of two correlations ( $\propto m^{2Q-1}$ ) is in their Lemma 21 and the exponent  $2Q$  in their total sample count  $N_1 T_1 = \tilde{\Omega}(d^{2Q+1} r)$ , framed as a consequence of correlational information; their Remark 3 states the  $N_1$ – $T_1$  trade-off, which Prop. 3 recovers for a shrinking readout and which fails for a readout pinned at order one (Prop. 2, Fig. 3). Nishikawa et al. (2025) show that softmax’s label nonlinearity lowers the inference-time context length to the generative exponent while one-step pretraining stays at  $d^{\Theta(\text{ie})}$ . Gu et al. (2026) derive, by the replica method for trained models with  $k^* = 1$ , the kernel-vs-feature context-

length scalings (their feature learner has a single transition at  $N = \Theta(d)$ , the same scaling as our  $d^*(N)$  but for inference, not for feature emergence) and the static shrinkage  $\Gamma^*$  (their eqs. (297)–(298)); they list jointly trained readouts and  $k^* \geq 2$  as open, which is where our results sit. Kim & Suzuki (2024) note that at finite sample size the ICL landscape may acquire a spurious minimum at the origin (their App. C.4), a qualitative precursor of Prop. 2. Ren et al. (2025) treat the additive model in-weight for even links with  $k^* > 2$  with a 2-homogeneous student; our tied protocol is theirs, the norm-independence of the directional dynamics is their Lemma B.1, and  $k^* = 2$  is outside their assumptions (the companion work Ben Arous et al. (2025) treats the quadratic case in-weight with a student narrower than  $d$ , sample complexity  $d \text{ polylog } d$ , and notes that the  $k \geq 3$  decoupling argument does not apply at  $k = 2$ ). The doubling of the exponent by a fast link is in Bietti et al. (2023, §5) and Bietti et al. (2022); two-timescale readout dynamics in two-layer networks are analysed by Berthier et al. (2023), and Ren et al. (2025, Cor. 2.3) give an in-weight scaling law in which the frequency ordering is also steepened by the learning-rate schedule. Zhang et al. (2024) prove, for linear regression, that the task mean is learned in-weight by the MLP and the covariance in-context. Skill-based accounts of emergence (Michaud et al., 2023; Nam et al., 2024; Arora & Goyal, 2023; Okawa et al., 2023) are in-weight; our composition result is the label-linear in-context counterpart. Compositions that are not output-additive need extra structure (Kobayashi et al., 2024; Wang et al., 2026); He et al. (2024) show that even a label-linear in-context algorithm is not free when the network must discover it against an in-weight memoriser, whereas in (1) linearity in the labels is built in.

## 7. Limitations

The theory is for one feature per skill and  $k^* = 2$ ; the SGD threshold is soft; the many-skill statements rest on decoupling, which fails for rare skills when no feature starts closest to them (2/10 seeds in Exp. 3); the Gaussian closed form for compositional accuracy is wrong; the transformer results are observational and were pre-registered with predictions that failed in both directions. All predictions, outcomes and deviations (including two algebra slips corrected after pre-registration and two ill-posed estimators) are listed verbatim in the appendix.

**Reproducibility and disclosure.** Code, pre-registration document and all runs: [repo]. AI assistance was used for literature search, implementation, derivation checking and drafting; every derivation was verified against the Monte-Carlo drift, and every citation against the full text.

## References

- Arora, S. and Goyal, A. A theory for emergence of complex skills in language models. *arXiv preprint arXiv:2307.15936*, 2023.
- Ben Arous, G., Gheissari, R., and Jagannath, A. Online stochastic gradient descent on non-convex losses from high-dimensional inference. *Journal of Machine Learning Research*, 22(106):1–51, 2021.
- Ben Arous, G., Erdogdu, M. A., Vural, N. M., and Wu, D. Learning quadratic neural networks in high dimensions: SGD dynamics and scaling laws. *arXiv preprint arXiv:2508.03688*, 2025.
- Berthier, R., Montanari, A., and Zhou, K. Learning time-scales in two-layers neural networks. *arXiv preprint arXiv:2303.00055*, 2023.
- Bietti, A., Bruna, J., Sanford, C., and Song, M. J. Learning single-index models with shallow neural networks. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2210.15651.
- Bietti, A., Bruna, J., and Pillaud-Vivien, L. On learning Gaussian multi-index models with gradient flow. *arXiv preprint arXiv:2310.19793*, 2023.
- Gu, H., Xu, Y., and Zdeborová, L. In-context learning of single-index targets: Comparing kernel and feature learners. *arXiv preprint arXiv:2610.01712*, 2026.
- He, T., Doshi, D., Das, A., and Gromov, A. Learning to grok: Emergence of in-context learning and skill composition in modular arithmetic tasks. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2406.02550.
- Kim, J. and Suzuki, T. Transformers learn nonlinear features in context: Nonconvex mean-field dynamics on the attention landscape. In *Proceedings of the 41st International Conference on Machine Learning*, 2024. arXiv:2402.01258.
- Kobayashi, S., Schug, S., Akram, Y., Redhardt, F., von Oswald, J., Pascanu, R., Lajoie, G., and Sacramento, J. When can transformers compositionally generalize in-context? In *ICML 2024 Workshop on Next Generation of Sequence Modeling Architectures*, 2024. arXiv:2407.12275.
- Michaud, E. J., Liu, Z., Girit, U., and Tegmark, M. The quantization model of neural scaling. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2303.13506.
- Nam, Y., Fonseca, N., Lee, S. H., Mingard, C., and Louis, A. A. An exactly solvable model for emergence and scaling laws in the multitask sparse parity problem. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2404.17563.

- Nishikawa, N., Song, Y., Oko, K., Wu, D., and Suzuki, T. Nonlinear transformers can perform inference-time feature learning. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.
- Okawa, M., Lubana, E. S., Dick, R. P., and Tanaka, H. Compositional abilities emerge multiplicatively: Exploring diffusion models on a synthetic task. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2310.09336.
- Oko, K., Song, Y., Suzuki, T., and Wu, D. Pretrained transformer efficiently learns low-dimensional target functions in-context. In *Advances in Neural Information Processing Systems*, 2024. arXiv:2411.02544.
- Ren, Y., Nichani, E., Wu, D., and Lee, J. D. Emergence and scaling laws in SGD learning of shallow neural networks. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2504.19983.
- Wang, Z., Dang, X., Lee, J. D., and Lyu, K. The power of power law: Asymmetry enables compositional reasoning. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. arXiv:2604.22951.
- Zhang, R., Wu, J., and Bartlett, P. L. In-context learning of a linear transformer block: Benefits of the MLP component and one-step GD initialization. *arXiv preprint arXiv:2402.14951*, 2024.

## A. Derivations

**Hermite algebra.** For unit  $u, w$  with  $\langle u, w \rangle = m$  and  $z_u = \langle u, x \rangle$ ,  $z_w = \langle w, x \rangle$  jointly Gaussian with correlation  $m$ ,  $\mathbb{E}[\sigma_j(z_u)\sigma_k(z_w)] = \delta_{jk}m^k$ . For  $\sigma = \sigma_2 = (z^2 - 1)/\sqrt{2}$ :  $\sigma_2^2 = \sqrt{6}\sigma_4 + 2\sqrt{2}\sigma_2 + 1$ , hence  $V(m) = \mathbb{E}[\sigma_2(z_u)^2\sigma_2(z_w)^2] = 6m^4 + 8m^2 + 1$  and  $g(m) = m^2$ . For  $\sigma_1 = z$ :  $g = m$ ,  $V = 1 + 2m^2$ .

**Population loss (Prop. 1).** With  $A = \frac{1}{N} \sum_i y_i \sigma(\langle w, x_i \rangle)$  and  $y_i = c \sigma(\langle v, x_i \rangle)$ ,  $\mathbb{E}[A | c] = c g(m)$  and  $\text{Var}[A | c] = c^2(V(m) - g(m)^2)/N$ . Since  $q$  is independent of the context,  $\mathbb{E}[y_q \hat{y}_q] = \Gamma \mathbb{E} c^2 g(m)^2$  and  $\mathbb{E}[\hat{y}_q^2] = \Gamma^2 g(1) \mathbb{E} c^2 (g^2 + (V - g^2)/N)$ , giving (2).

**Drift (Prop. 2).** For  $\sigma_2$ ,  $\Gamma = \gamma$ ,  $s^2 = 1$ :  $L_A = 1 - 2\gamma m^4 + \gamma^2[(1 - \frac{1}{N})m^4 + (1 + 8m^2 + 6m^4)/N]$ , so  $-\partial_m L_A = (8\gamma - 4\gamma^2(1 - \frac{1}{N}) - 24\gamma^2/N)m^3 - 16\gamma^2 m/N$ , which is negative for  $m < m^*$  with  $m^{*2} = 16\gamma/(N(8 - 4\gamma(1 - \frac{1}{N}) - 24\gamma/N))$ . (An earlier draft of this derivation dropped the  $24\gamma^2/N$  term and had a factor 2 in the cubic coefficient; the closed-form loss, which the Monte-Carlo drift check verifies directly, was unaffected.)

**General  $\sigma_k$ .** The normalised second Hermite coefficient of  $\sigma_k^2$  is  $\beta_2 = \sqrt{2}k$  (checked by quadrature for  $k \leq 4$ ), so the finite-context repulsion is  $-\Gamma^2 \beta_2^2 2m/N = -4k^2 \Gamma^2 m/N$  to leading order while the attraction is  $2\Gamma(2 - \Gamma g(1))k m^{2k-1}$ : the origin is drift-stable below  $m^{*2(k-1)} \approx 2k\Gamma/((2 - \Gamma)N)$  for every  $k \geq 2$  (for  $k = 1$  the two terms have the same power and the repulsion is a constant fraction). At  $k = 3$ ,  $\Gamma = 0.1$ ,  $N = 128$  the trap sets in at  $d^* \approx 17$  (against  $\approx 607$  at  $k = 2$ ), consistent with the negative Monte-Carlo drift observed for  $k^* = 3$  at  $d = 16$ . The step size  $\eta \propto d^{-2}$  is the critical scaling only for  $k^* = 2$ .

**Tied = pinned up to a time change.** The tied directional velocity equals the pinned velocity at  $\gamma = \rho$  divided by  $\rho$ ; with  $\dot{\rho} = -4\rho \partial_\rho L_A$  the pair  $(m, \rho)$  above threshold obeys  $m \propto t^{-2}$ ,  $\rho \propto t^{-1}$  (LSODA to  $t = 10^6$ ), and the moving threshold  $\rho^*(m) \approx Nm^2/2 \propto t^{-4}$  falls faster than  $\rho$ : the population flow traps permanently.

**Escape-time scalings in Prop. 3.** (a) With  $\Gamma = \gamma$  fixed and  $m_0 > m^*$  the drift near the origin is  $\approx c_\gamma m^{2k^*-1}$  with  $c_\gamma = \gamma(8 - 4\gamma)k^* \alpha_{k^*}^4/2$  for  $k^* = 2$ , so  $\tau_{\text{esc}} \asymp m_0^{-(2k^*-2)}/c_\gamma$  up to the  $N$ -dependent correction  $-16\gamma^2 m/N$ , which saturates as  $N \rightarrow \infty$ . (b) Along  $\Gamma^*$ , for  $g^2 \ll V/N$ ,  $L_A(m, \Gamma^*) \approx s^2[1 - Ng(m)^4/(g(1)V(m))]$  and  $V(m) \rightarrow \beta_0^2 = g(1)^2$  as  $m \rightarrow 0$ , so the drift is  $\approx s^2 N 4k^* \alpha_{k^*}^8 m^{4k^*-1}/g(1)^3$ :  $\tau_{\text{esc}} \asymp m_0^{-(4k^*-2)}/N$ , i.e.  $\tau_{\text{esc}} \propto d^{2k^*-1}/N$ —the multiplicative  $N$ - $\tau$  trade-off. The condition for this regime at the initial overlap is  $m_0^{2k^*} \ll V/N$ , i.e.  $d^{k^*} \gg N$ ; for  $d^{k^*} \ll N$  the signal-dominated form  $1 - g^2/g(1)$  applies and  $\tau_{\text{esc}} \asymp m_0^{-(2k^*-2)}$  is  $N$ -independent. (c) For the tied readout the attraction term  $4gg'$  carries no  $\rho$ , so  $\tau_{\text{esc}} \asymp m_0^{-(2k^*-2)}/(4k^* \alpha_{k^*}^4)$  independently of  $N$  as long as  $\rho V'/N \ll 4gg'$  along the trajectory, which holds from  $\rho_0 \ll \rho_{\text{trap}}(m_0, N)$  (Theorem A(iii)), which for  $k^* = 2$  reads  $\rho_0 \ll Nm_0^2/2$ ; for  $k^* = 3$ ,  $\rho_{\text{trap}} \propto m_0^4$  and  $\rho_0 = 0.01$  already violates it at  $N = 16$ ,  $d \geq 16$  (Exp. 9). At  $d = 512$ ,  $k^* = 2$  the three regimes give  $\tau \propto \{\infty \text{ or } N\text{-saturating}, N^{-1}, N^0\}$ , matching the integrated flows ( $\infty/771/386$ ;  $9.0 \cdot 10^4/2.2 \cdot 10^4/5.4 \cdot 10^3$ ;  $39/34/33$  for  $N = 32/128/512$ ).

**Population ODEs (Prop. 3).** Spherical gradient flow for the overlap and plain flow for the readout: free readout  $\dot{m} = -(1 - m^2)\partial_m L_A$ ,  $\dot{\Gamma} = -r \partial_\Gamma L_A$  with  $r = \eta_\Gamma/\eta$ ; tied readout  $\dot{m} = (1 - m^2)[4gg' - \rho(2(1 - \frac{1}{N})gg' + V'/N)]$ ,  $\dot{\rho} = -4\rho \partial_\rho L_A$ . The chain rule through  $\hat{u} = u/\|u\|$  gives the directional velocity a factor  $1/\|u\|^2$  that cancels the  $\rho = \|u\|^2$  carried by the attraction term but not the  $\rho$  carried by the repulsion. Escape times  $T_{0.5}$  to  $m = 0.5$  from  $m_0 = c d^{-1/2}$  were integrated with an adaptive step (1% relative change per step) and the exponent read off as  $\kappa_{\text{eff}} = 2 - \partial \log T / \partial \log m_0$  over  $c \in \{0.5, 0.7, 1, 1.4\}$ , exactly as for the SGD data.

**Kramers estimate.**  $D_{\text{eff}} = \eta \text{sd}^2/2$  with the per-step standard deviation of the  $m$ -component of the batch gradient measured by Monte Carlo (0.33–0.37 at  $B = 32$ ); the estimate ignores the spherical correction term, which is of the same order as the quoted  $d = 32$  ratio, so only the  $d = 64$  value ( $\approx 20$ ) is meaningful.

**Monte-Carlo drift check.**  $d = 16$ ,  $N = 128$ ,  $\gamma = 0.1$ ,  $a = 1$ ,  $5 \cdot 10^4$  batches of 64 prompts per point. Ratios MC/formula at  $m = 0.2$  and  $0.4$ : Model B  $k^* = 1, 2, 3$ : 0.998–1.00; Model A  $k^* = 1$ : 1.000;  $k^* = 2$ : 0.972, 0.998;  $k^* = 3$ :  $0.88 \pm 0.76$  (drift  $\approx 0$ ), 1.02. Model A  $k^* = 3$  at  $\gamma = 0.1$ ,  $N = 128$  has negative drift for  $m \leq 0.2$ .

## B. Theorems and proofs

All closed forms in this section are checked numerically by `scripts/verify_theorems.py` (61 checks: thresholds, the exact escape time against LSODA, the time-change identity, the  $t^{-2}/t^{-1}$  asymptotics, Theorem B at  $d$  up to  $6.5 \cdot 10^4$ , the corner exponents).

**Standing notation.**  $\sigma = \sigma_k$ ,  $k \geq 2$ , so  $\alpha_k = 1$ ,  $g(m) = m^k$ ,  $g(1) = 1$ . Write  $\sigma_k^2 = \sum_{i=0}^k \beta_{2i} \sigma_{2i}$ ; the linearisation formula  $\text{He}_k^2 = \sum_j j! \binom{k}{j}^2 \text{He}_{2k-2j}$  gives  $\beta_{2i}^2 = ((k-i)! \binom{k}{i})^2 (2i)!/k!^2 \geq 0$ , in particular  $\beta_0^2 = 1$ ,  $\beta_2^2 = 2k^2$ ,  $\beta_{2k}^2 = \binom{2k}{k}$ , and  $V(m) = \sum_{i=0}^k \beta_{2i}^2 m^{2i}$ ,  $V'(1) = 4k \mathbb{E}[\sigma_k^2 \sigma_{k-1}^2]$  ( $= 40$  for  $k = 2$ ). Put  $G = g^2 = m^{2k}$ ,  $D(m) = (1 - \frac{1}{N})m^{2k} + V(m)/N$ , so that

$$L(m, \Gamma) = s^2[1 - 2\Gamma G + \Gamma^2 D], \quad \partial_m L = s^2(\Gamma^2 D' - 2\Gamma G'), \quad \partial_\Gamma L = 2s^2(\Gamma D - G), \quad \Gamma^*(m) = G/D.$$

Two elementary facts are used throughout:  $D \geq V(0)/N = 1/N$ , and  $mV' \leq 2kV$  (hence  $mD' \leq 2kD$ ), because  $V$  has degree  $2k$  and nonnegative coefficients. Flows: pinned  $\dot{m} = -(1 - m^2)\partial_m L(m, \gamma)$ ; free  $\dot{m} = -(1 - m^2)\partial_m L$ ,  $\dot{\Gamma} = -r\partial_\Gamma L$ ; tied  $\dot{m} = -(1 - m^2)\partial_m L(m, \rho)/\rho$ ,  $\dot{\rho} = -4\rho\partial_\rho L(m, \rho)$ . The interval  $(0, 1)$  is invariant for  $m$  in all three ( $g'(0) = V'(0) = 0$  for  $k \geq 2$ ),  $\tau_{\text{esc}} = \inf\{t : m(t) \geq \frac{1}{2}\}$ , and  $t$  is flow time ( $\eta$ -steps).

**Theorem A** (Finite-context trap; population flow,  $\sigma = \sigma_k$ ,  $k \geq 2$ ). (i) Pinned readout  $\Gamma \equiv \gamma > 0$ .  $\dot{m} = s^2\gamma(1 - m^2)mQ_\gamma(m^2)$  with

$$Q_\gamma(x) = 2k(2 - \gamma(1 - \frac{1}{N}))x^{k-1} - \frac{\gamma}{N} \sum_{i=1}^k 2i\beta_{2i}^2 x^{i-1}.$$

Let  $\gamma_1 = 2N/(N - 1 + V'(1)/2k)$ . If  $\gamma < \gamma_1$ ,  $Q_\gamma$  has exactly one positive root  $x^*$ , and  $x^* \in (0, 1)$ ; with  $m^* = \sqrt{x^*}$ ,  $m(t) \downarrow 0$  (exponentially, at rate  $\rightarrow 4k^2 s^2 \gamma^2/N$ ) if  $m_0 < m^*$ , and  $m(t) \uparrow 1$  with  $\tau_{\text{esc}} < \infty$  if  $m^* < m_0 < \frac{1}{2}$ . If  $\gamma \geq \gamma_1$ ,  $m(t) \rightarrow 0$  from every  $m_0 \in (0, 1)$ . As  $N \rightarrow \infty$  at fixed  $\gamma \in (0, 2)$ ,  $m^{*2(k-1)} = \frac{2k\gamma}{(2-\gamma)N}(1 + O(N^{-1/(k-1)}))$ . For  $k = 2$ :  $\gamma_1 = \frac{2N}{N+9}$  and  $m^{*2} = \frac{4\gamma}{2N-(N+5)\gamma}$ .

(ii) Exact escape time,  $k = 2$ . Let  $C = \gamma(8 - 4\gamma - 20\gamma/N)$ ,  $B = 16\gamma^2/N$  ( $C > B \iff \gamma < \gamma_1$ ) and  $m^{*2} < m_0^2 < \frac{1}{4}$ . Then

$$\tau_{\text{esc}} = \frac{F(\frac{1}{4}) - F(m_0^2)}{2s^2}, \quad F(u) = -\frac{\ln u}{B} - \frac{\ln(1-u)}{C-B} + \frac{C \ln(Cu-B)}{B(C-B)}.$$

Hence  $2s^2 C m_0^2 \tau_{\text{esc}} = 1 + O(m_0^2 \ln \frac{1}{m_0} + m^{*2}/m_0^2)$ , and  $\tau_{\text{esc}} \sim \frac{C}{2s^2 B(C-B)} \ln \frac{1}{m_0^2 - m^{*2}}$  as  $m_0 \downarrow m^*$ .

(iii) Tied readout. (a)  $\dot{m}_{\text{tied}}(m, \rho) = \dot{m}_{\text{pinned}}(m; \gamma=\rho)/\rho$  identically. Hence, if  $(m, \rho)(t')$  is the tied solution and  $t = \int_0^{t'} du/\rho(u)$ , then  $m(t)$  solves the pinned flow with the time-dependent value  $\gamma(t) = \rho$  (i.e.  $dt' = \rho dt$ ); for  $\rho$  frozen at  $\rho_0$ ,  $\tau_{\text{tied}} = \rho_0 \tau_{\text{pinned}}(\gamma=\rho_0)$ . (b) Let  $m_0 < \frac{1}{2}$ ,  $D_0 = D(m_0)$  and  $a = k^2(k-1)(1 - m_0^2)/(ND_0)$ . If  $a > 1$  and

$$\rho_0 > \rho_{\text{trap}}(m_0) := \frac{k(k-1)m_0^{2k-2}}{D_0(a-1)}, \quad (3)$$

the tied flow is trapped for ever:  $m(t) \leq m_0$  for all  $t$ ,  $m(t) \rightarrow 0$ , and, as  $t \rightarrow \infty$ ,  $\rho(t)t \rightarrow N/(8s^2)$  and  $m(t)t^{k^2/2} \rightarrow c \in (0, \infty)$ ; for  $k = 2$ ,  $m \propto t^{-2}$ ,  $\rho \propto t^{-1}$ . Since  $\rho^*(m) := 2G'/D' \asymp Nm^{2k-2}/k$ ,  $\rho/\rho^*(m) \rightarrow \infty$ : the threshold recedes faster than  $\rho$  decays.

(iv) For every number  $B$  of prompts per step the expected minibatch gradient is  $\nabla L$ ; the population flows above, and hence  $m^*(\gamma, N, k)$ ,  $\gamma_1$  and (ii)–(iii), do not depend on  $B$ .

**Lemma A.1** (Monotone signal-to-noise).  $R(m) = G^2/D$  is strictly increasing on  $(0, 1]$ , and  $\rho^*(m) = 2G'/D'$  satisfies  $\rho^* > \Gamma^*$  and is strictly increasing.

*Proof.*  $mR'/R = 4k - mD'/D \geq 2k > 0$ .  $\rho^* > \Gamma^* \iff 2G'D > GD' \iff 4kD > mD'$ . Finally  $\rho^* = 4k/(2k(1 - \frac{1}{N}) + N^{-1} \sum_i 2i\beta_{2i}^2 m^{2i-2k})$  and each  $m^{2i-2k}$ ,  $i < k$ , is strictly decreasing.  $\square$

*Proof of Theorem A.* (i) Insert  $G' = 2km^{2k-1}$  and  $V'(m)/m = \sum_{i \geq 1} 2i\beta_{2i}^2 m^{2i-2}$  in  $-\partial_m L$ . The coefficients of  $Q_\gamma$  of degree  $< k-1$  are strictly negative ( $\beta_{2i}^2 > 0$ ), and  $Q_\gamma(0) = -4k^2\gamma/N < 0$ . By Descartes' rule  $Q_\gamma$  has at most one positive root; it has one iff the leading coefficient  $2k[2 - \gamma(N-1 + \beta_{2k}^2)/N]$  is positive, and the root lies in  $(0, 1)$  iff  $Q_\gamma(1) = 2k[2 - \gamma(N-1 + V'(1)/2k)/N] > 0$ , i.e.  $\gamma < \gamma_1$  (this implies the former, as  $V'(1)/2k > \beta_{2k}^2$ ). Then  $Q_\gamma < 0$  on  $(0, x^*)$  and  $> 0$  on  $(x^*, \infty)$ ; if  $\gamma \geq \gamma_1$ ,  $Q_\gamma < 0$  on  $(0, 1)$ . The one-dimensional phase portrait follows: on  $[m_0, \frac{1}{2}] \subset (m^*, 1)$ ,  $\dot{m} \geq c > 0$ ; on  $(0, m_0] \subset (0, m^*)$ ,  $\dot{m}/m \leq -c' < 0$  and  $\dot{m}/m \rightarrow s^2\gamma Q_\gamma(0)$ . For the asymptotics, the root equation  $2k(2 - \gamma(1 - \frac{1}{N})) x^{*k-1} = \frac{\gamma}{N} \sum_i 2i\beta_{2i}^2 x^{*i-1}$  forces  $x^* \rightarrow 0$  and gives  $x^{*k-1} = \frac{4k^2\gamma}{2k(2-\gamma)N}(1 + O(x^* + N^{-1}))$  with  $x^* = O(N^{-1/(k-1)})$ . For  $k=2$ ,  $Q_\gamma(x) = (Cx - B)/\gamma$ .

(ii) With  $u = m^2$ ,  $\dot{u} = 2s^2u(1-u)(Cu - B)$ , and  $\frac{1}{u(1-u)(Cu-B)} = -\frac{1}{Bu} + \frac{1}{(C-B)(1-u)} + \frac{C^2}{B(C-B)(Cu-B)}$ ; integrate from  $u_0$  to  $\frac{1}{4}$ . Expanding  $F$  in  $\varepsilon = B/(Cu_0)$  gives  $2s^2(F(\frac{1}{4}) - F(u_0)) = \frac{1}{Cu_0}(1 + O(\varepsilon)) + O(C^{-1} \ln \frac{1}{u_0})$ ; the logarithmic singularity at  $u_0 = B/C$  is the last term of  $F$ .

(iii)(a)  $\partial_m L(m, \gamma) = s^2\gamma(\gamma D' - 2G')$ , so  $\dot{m}_{\text{pinned}}/\gamma = s^2(1 - m^2)(2G' - \gamma D')$ , which is the tied velocity  $-(1 - m^2)\partial_m L(m, \rho)/\rho$  at  $\gamma = \rho$  (the  $1/\rho$  is the chain rule through  $w = u/\|u\|$ ,  $\rho = \|u\|^2$ ; cf. Ren et al., 2025, Lemma B.1). Reparametrising time by  $dt = dt'/\rho$  gives the claim.

(b) Write  $z = m^{-(2k-2)}$ . While  $m \leq \bar{m}$  ( $\bar{m} \downarrow m_0$  at the end), using  $1 - m^2 \leq 1$  in the attraction,  $1 - m^2 \geq 1 - \bar{m}^2$  and  $V'(m)/m \geq 4k^2$  in the repulsion, dropping the term  $-2k(1 - \frac{1}{N})\rho m^{2k-2}$ , and using that  $D$  is increasing,

$$\frac{d \ln m}{dt} \leq s^2 \left[ 4km^{2k-2} - (1 - \bar{m}^2) \frac{4k^2\rho}{N} \right], \quad \dot{\rho} = 8s^2\rho(G - \rho D) \geq -8s^2D(\bar{m})\rho^2,$$

so  $\rho \geq \rho_0/E$  with  $E(t) = 1 + 8s^2D(\bar{m})\rho_0 t$ , and  $\dot{z} \geq -(2k-2)4ks^2 + (2k-2)(1 - \bar{m}^2) \frac{4k^2s^2}{N} \rho z$ . Comparison with the linear equation gives  $z(t) \geq (z_0 - K)E^a + KE$  with  $K = k(k-1)/(\rho_0 D(\bar{m})(a-1))$  (and  $a$  evaluated at  $\bar{m}$ ). By (3)  $z_0 > K$  for  $\bar{m}$  close to  $m_0$ , so the right side is increasing in  $E \geq 1$  and  $z(t) \geq z_0$ :  $m \leq m_0 < \bar{m}$  for all  $t$  (continuity argument),  $m \rightarrow 0$ , and  $\rho z \geq \rho_0(z_0 - K)E^{a-1} \rightarrow \infty$ . Restarting the argument at a late time  $t_0$  (where  $ND(m(t_0)) \rightarrow 1$ , so (3) holds with  $a$  arbitrarily close to  $k^2(k-1)$ ) gives  $m^{2k-2} \leq Ct^{-a'}$  for every  $a' < k^2(k-1)$ . Hence  $m^2$  and  $G/\rho$  are integrable in  $t$ , so  $\frac{d}{dt} \frac{1}{\rho} = 8s^2(D - G/\rho) = \frac{8s^2}{N} + (\text{integrable})$  gives  $\frac{1}{\rho} = \frac{8s^2 t}{N} + c_1 + o(1)$ . Then  $\frac{d \ln m}{dt} = -\frac{4k^2s^2}{N}\rho(1 + O(m^2)) + O(m^{2k-2}) = -\frac{k^2}{2t} + (\text{integrable})$ , so  $mt^{k^2/2}$  converges to a positive limit.

(iv) The per-prompt losses in a step are i.i.d. with mean  $L$ ; the gradient of their average has mean  $\nabla L$  for every  $B$  (differentiation under  $\mathbb{E}$  is justified by polynomial moments), and  $B$  enters only the covariance, which is  $\propto 1/B$ .  $\square$

**Remark 2** (The trap boundary is  $c_k\rho^*$ , not  $\rho^*$ ). Condition (3) is sharp as  $m_0 \rightarrow 0$ : since  $ND_0 \rightarrow 1$  and  $a \rightarrow k^2(k-1)$ , it reads  $\rho_0 > c_k Nm_0^{2k-2}/k(1 + o(1))$  with  $c_k = 1 + 1/(k^3 - k^2 - 1)$  ( $c_2 = \frac{4}{3}$ ,  $c_3 = \frac{18}{17}$ ), and in the small- $m$  system  $\dot{m} = s^2m(4km^{2k-2} - 4k^2\rho/N)$ ,  $\dot{\rho} = -8s^2\rho^2/N$  (exactly  $z = (z_0 - K)E^a + KE$  with  $a = k^2(k-1)$ ,  $ND_0 = 1$ )  $z$  reaches 0 in finite time, i.e. the flow escapes, for every  $\rho_0 < c_k Nm_0^{2k-2}/k$ . So for  $\rho^*(m_0) < \rho_0 < c_k\rho^*(m_0)$  the overlap first decreases and then escapes: “ $\rho_0 > \rho^*(m_0) \Rightarrow \text{trap}$ ” is false. LSODA bisection of the exact flow gives the boundary  $\rho_0/\rho^*(m_0) = 1.3333$  ( $k=2$ ,  $m_0=10^{-3}$ ), 1.0589 ( $k=3$ ,  $m_0=10^{-2}$ ) and 1.3735 at the E5 cell ( $d=64$ ,  $N=16$ ,  $\rho^* = 0.1155$ ), where (3) gives  $\rho_{\text{trap}} = 0.178$ :  $\rho_0 = 0.3$  is provably trapped, and  $\rho_0 = 0.1 < \rho^*$  escapes. The escape half is proved only for the small- $m$  system; for the exact flow it is a numerical statement.

**Theorem B** (Free readout, fixed  $N$ ,  $d \rightarrow \infty$ ). Assume (B1)  $\sigma = \sigma_k$ ,  $k \geq 2$ ; (B2)  $N \geq 2$ ,  $r > 0$ ,  $\Gamma_0 > 0$ ,  $s > 0$  and  $\mu_0 > 0$  fixed; (B3)  $m_0 = \mu_0 d^{-1/2}$ ,  $d \rightarrow \infty$ . Then

$$\tau_{\text{esc}} = (1 + o(1)) \exp\left(\frac{(4k-2)\beta_2^2\Gamma_0^2}{2r}\right) \frac{\mu_0^{-(4k-2)} d^{2k-1}}{4k(4k-2)\alpha_k^8 s^2 N}, \quad \beta_2^2 = 2k^2, \alpha_k = 1.$$

In particular the escape exponent is  $4k$  for every finite  $N$ , and  $N$  enters only as the prefactor  $1/N$ .

*Proof.* Rescale  $t \mapsto s^2 t$  (this maps the flow with  $s$  to the flow with  $s=1$  and the same  $r$ ), so  $s=1$ . Put  $m_1 = m_0 e^{-k^2\Gamma_0^2/r}$  and note  $\frac{(4k-2)\beta_2^2\Gamma_0^2}{2r} = (4k-2)\frac{k^2\Gamma_0^2}{r}$ , so the claim is  $\tau_{\text{esc}} \sim \tau_B := m_1^{-(4k-2)}/(4k(4k-2)N)$ . The equations are

$$\frac{d \ln m}{dt} = (1 - m^2) \left[ 4k\Gamma m^{2k-2} \left( 1 - \frac{\Gamma}{2} \left( 1 - \frac{1}{N} \right) \right) - \frac{\Gamma^2 V'(m)}{Nm} \right], \quad \dot{\Gamma} = -2rD(m)(\Gamma - \Gamma^*(m)),$$

with  $V'(m)/m = 4k^2(1 + O(m^2))$ ,  $D = N^{-1}(1 + O(m^2))$ ,  $\Gamma^* = Nm^{2k}(1 + O(m^2 + Nm^{2k}))$ ; constants below depend on  $(k, N, r, \Gamma_0, \mu_0)$  only.

*Phase 1*,  $0 \leq t \leq t_1 = \frac{(k-1)N}{4r} \ln d$  (readout transient). While  $m \leq 2m_0$ ,  $\dot{\Gamma} = -\frac{2r}{N}(1 + O(d^{-1}))\Gamma + O(d^{-k})$ , and  $\Gamma \geq \Gamma_0 d^{-(k-1)/2}/2 \gg Nd^{-k}$ , so  $\Gamma(t) = \Gamma_0 e^{-2rt/N}(1 + o(1))$  uniformly. Integrating the  $m$ -equation,  $\ln \frac{m(t_1)}{m_0} = -\frac{4k^2}{N} \int_0^{t_1} \Gamma^2(1 + o(1)) + O(m_0^{2k-2} \int (\Gamma + \Gamma^2)) = -\frac{k^2 \Gamma_0^2}{r} + o(1)$ , because  $\int_0^{t_1} \Gamma^2 = \frac{N\Gamma_0^2}{4r}(1 + o(1))$  and  $m_0^{2k-2} \int (\Gamma + \Gamma^2) = O(d^{1-k}N/r)$ ; the same bounds keep  $m \leq m_0(1 + o(1))$ , closing the bootstrap.

*Phase 2*,  $t_1 \leq t \leq t_2 = t_1 + \frac{(k+1)N}{2r} \ln d$  (relaxation onto  $\Gamma^*$ ). Now  $\Gamma \leq 2\Gamma_0 d^{-(k-1)/2}$ , so  $|\frac{d \ln m}{dt}| = O(d^{1-k}\Gamma + \Gamma^2)$  integrates to  $o(1)$ :  $m(t_2) = m_1(1 + o(1))$ . With  $e = \Gamma - \Gamma^*$ ,  $\dot{e} = -2rDe - \Gamma^* \dot{m}$  and  $D \geq 1/N$ ,  $|e(t_2)| \leq d^{-(k+1)}\Gamma_0 + \frac{N}{2r} \sup |\Gamma^* \dot{m}| = o(d^{-k}) = o(\Gamma^*(m(t_2)))$ .

*Phase 3*, adiabatic, until  $t_3$ , the first time  $m = m_\delta$ . Fix  $\delta \in (0, \frac{1}{4})$  and choose  $m_\delta$  (independent of  $d$ ) so small that for  $m \leq m_\delta$  the  $O(\cdot)$  terms above are  $\leq \delta$  and  $\epsilon_\delta := 32k^2 Nm_\delta^{4k-2} \leq \delta r/N$ . Write  $\Gamma = \Gamma^*(1 + e')$ . While  $|e'| \leq 1$ ,  $0 < \frac{d \ln m}{dt} = 4kNm^{4k-2}(1 + e')(1 + O(\delta))$  and, as  $0 \leq \frac{d \ln \Gamma^*}{d \ln m} = 2k - \frac{mD'}{D} \leq 2k$ ,  $|\frac{d \ln \Gamma^*}{dt}| \leq \epsilon_\delta$ ; and  $\dot{e}' = -(2rD + \frac{d \ln \Gamma^*}{dt})e' - \frac{d \ln \Gamma^*}{dt}$ . At  $|e'| = \delta$  the right side has the sign of  $-e'$ , since  $(\frac{2r}{N} - \epsilon_\delta)\delta > \epsilon_\delta$ . Hence  $|e'| \leq \delta$  on  $[t_2, t_3]$ , and  $t_3 - t_2 = (1 + O(\delta)) \frac{m(t_2)^{-(4k-2)} - m_\delta^{-(4k-2)}}{4k(4k-2)N} = (1 + O(\delta) + o(1)) \tau_B$ .

*Phase 4*,  $t \geq t_3$ .  $(m, \Gamma)(t_3) \in K_\delta = \{m_\delta\} \times [(1 - \delta)\Gamma^*, (1 + \delta)\Gamma^*]$ , a compact set on which  $L < 1 = L(0, 0)$  (since  $L = 1 - \Gamma G(2 - (1 + e'))$ ).  $L$  is a Lyapunov function,  $\frac{dL}{dt} = -(1 - m^2)(\partial_m L)^2 - r(\partial_\Gamma L)^2$ , trajectories are bounded, and the only equilibria in  $[0, 1] \times [0, \infty)$  are  $(0, 0)$  and, for  $m \in (0, 1)$ , points with  $\Gamma = \Gamma^*(m)$  and  $\frac{d}{dm} L(m, \Gamma^*(m)) = -R'(m) = 0$  (envelope theorem), which Lemma A.1 excludes. By LaSalle every trajectory from  $K_\delta$  has  $m \rightarrow 1$ ; the sets  $\{x : m \text{ exceeds } \frac{1}{2} \text{ before time } T\}$  are open and cover  $K_\delta$ , so the remaining time is  $\leq T_\delta < \infty$ , independent of  $d$ .

Phases 1, 2 and 4 last  $O(N \ln d/r) + T_\delta = o(\tau_B)$ , as  $\tau_B \asymp d^{2k-1}$ . Thus  $\limsup |\tau_{\text{esc}}/\tau_B - 1| \leq C\delta$  for every  $\delta$ .  $\square$

**Proposition 4** (Joint scaling; proof sketch). Let  $N = \lambda d^k$ ,  $r = \tilde{r}d$ ,  $m_0 = \mu_0 d^{-1/2}$ ,  $\Gamma_0 \in (0, 2)$ ,  $s = 1$ , and put  $\mu = \sqrt{d}m$ ,  $\theta = t d^{1-k}$ . As  $d \rightarrow \infty$  the flow converges, uniformly on compacts of  $\{\mu > 0\}$  with an  $O(1/d)$  error, to the  $d$ -free system

$$\frac{d\mu}{d\theta} = 4k\Gamma(1 - \frac{\Gamma}{2})\mu^{2k-1}, \quad \frac{d\Gamma}{d\theta} = 2\tilde{r}\left[\mu^{2k} - \Gamma\left(\mu^{2k} + \frac{1}{\lambda}\right)\right], \quad (4)$$

whose solution blows up ( $\mu \rightarrow \infty$ ) at a finite  $T(\mu_0; \lambda, \tilde{r}, \Gamma_0)$ , and  $\tau_{\text{esc}} d^{1-k} \rightarrow T$ . (The finite- $N$  repulsion is  $O(1/d)$  in this scaling.) With  $\kappa := 2 - \partial \ln T / \partial \ln \mu_0$  (the secant exponent of the experiments is a finite-difference estimate of it): (a)  $\lambda T \rightarrow \mu_0^{-(4k-2)} / (4k(4k-2))$  as  $\lambda \rightarrow 0$  ( $\tilde{r}, \Gamma_0, \mu_0$  fixed):  $\kappa \rightarrow 4k$ ; (b) at  $\lambda = \infty$  and fixed  $\Gamma_0 \in (0, 2)$ ,  $\mu_0^{2k-2} T \rightarrow 1 / ((2k-2)2k\Gamma_0(2 - \Gamma_0))$  as  $\mu_0 \rightarrow 0$ :  $\kappa \rightarrow 2k$ ; (c) at  $\lambda = \infty$ ,  $\Gamma_0 = 0$  (readout growth-limited),  $\mu_0^{2k-1} T \rightarrow \frac{\pi}{2} \frac{(2k-3)!!}{(2k-2)!!} / (2\sqrt{2k\tilde{r}})$  as  $\sqrt{\tilde{r}}\mu_0 \rightarrow 0$ :  $\kappa \rightarrow 2k + 1$ .

*Sketch.* Substituting,  $d^{k-\frac{1}{2}}\dot{m} = (1 - \frac{\mu^2}{d})[2k\Gamma(2 - \Gamma(1 - \frac{1}{N}))\mu^{2k-1} - \frac{\Gamma^2}{\lambda} \sum_i 2i\beta_{2i}^2 \mu^{2i-1} d^{-i}]$  and  $d^{k-1}\dot{\Gamma} = 2\tilde{r}[\mu^{2k} - \Gamma((1 - \frac{1}{N})\mu^{2k} + V(\mu d^{-1/2})/\lambda)]$ , which is (4) up to  $O((1 + \mu^{2k+2})/d)$ . In (4),  $\Gamma \leq \max(\Gamma_0, 1) < 2$  (at  $\Gamma \geq 1$ ,  $\dot{\Gamma} < 0$ ) and  $\Gamma \geq c := \min(\Gamma_0, \Gamma_\infty^*(\mu_0))$  with  $\Gamma_\infty^* = \mu^{2k}/(\mu^{2k} + \lambda^{-1})$  increasing along the (increasing)  $\mu$ ; so  $\dot{\mu} \geq c'\mu^{2k-1}$  and  $\mu$  blows up in finite time. Split at  $\mu = M$ : up to  $M$ , continuous dependence gives convergence of the hitting time; after  $M$  both the limit and (by the same invariance bounds, the repulsion being  $O(M^{2-2k}/\lambda d)$ ) the finite- $d$  flow need time  $\leq CM^{2-2k}$ , which  $\rightarrow 0$  as  $M \rightarrow \infty$ . Corners: (a)  $\Gamma$  relaxes at rate  $2\tilde{r}/\lambda \rightarrow \infty$  onto  $\Gamma_\infty^* \approx \lambda\mu^{2k}$ , the transient moves  $\ln \mu$  by  $O(\Gamma_0\lambda/\tilde{r})$ , and  $\dot{\mu} = 4k\lambda\mu^{4k-1}(1 + o(1))$ . (b)  $d\Gamma/d\mu = \tilde{r}\mu(1 - \Gamma)/(2k\Gamma(1 - \Gamma/2))$ , so  $\Gamma = \Gamma_0 + O(\tilde{r}\mu_0^2 M^2)$  on  $[\mu_0, M\mu_0]$ , which carries all but  $O(M^{2-2k})$  of  $T$ . (c) For  $\Gamma \ll 1$ ,  $\Gamma^2 = \frac{\tilde{r}}{2k}(\mu^2 - \mu_0^2)$  is an approximate first integral, so  $\dot{\mu} = 4k\sqrt{\tilde{r}/2k} \sqrt{\mu^2 - \mu_0^2} \mu^{2k-1}$  and  $T = \mu_0^{1-2k} \int_1^\infty \frac{dx}{4k\sqrt{\tilde{r}/2k} x^{2k-1} \sqrt{x^2 - 1}}$ ; the region where  $\Gamma = O(1)$  costs  $O(\tilde{r}^{k-1}) = o(\mu_0^{1-2k}\tilde{r}^{-1/2})$ . The interchange of the corner limits with  $\partial/\partial \ln \mu_0$  (needed for  $\kappa$  rather than for  $T$ ) is not proved here; numerically the limits hold for the derivative as well (script).  $\square$

**Remark 3** (What is needed to transfer Theorem B to online SGD; heuristic). (I) Feature step. A drift  $\eta c m^{\kappa-1}$  beats the spherical renormalisation drag  $\asymp \eta^2 d \sigma_w^2 m/B$  at  $m \asymp d^{-1/2}$  iff  $\eta \lesssim cB d^{-\kappa/2} / \sigma_w^2$ : the critical scaling  $\eta \asymp d^{-\kappa/2}$  of Ben Arous et al. (2021), with  $\kappa = 2k$  (pinned, tied) or  $4k$  (free). Every  $w$ -gradient carries a factor  $\Gamma$ , so  $\sigma_w^2 \asymp \Gamma^2/N$  near

$m = 0$ , which relaxes the free-readout requirement once  $\Gamma \approx \Gamma^*$ . (2) Readout noise. The SGD readout fluctuates around  $\Gamma^*$  with stationary variance  $\text{Var}(\Gamma) \asymp \eta_{\Gamma}/B$  (relaxation rate and per-prompt variance both  $\asymp 1/N$ ), independent of  $N$ ; as  $L$  is quadratic in  $\Gamma$  this adds  $\text{Var}(\Gamma)D'(m)$  to the repulsion. Pathwise tracking,  $\text{Var}(\Gamma) \ll \Gamma^{*2} \asymp N^2 d^{-2k}$  at  $m \asymp d^{-1/2}$ , needs  $\eta_{\Gamma} \ll BN^2 d^{-2k}$ ; the mean  $m$ -drift alone needs only  $\eta_{\Gamma} \ll BN^2 d^{1-2k}/k$ . At  $\eta_{\Gamma} = r\eta_0 d^{-2}$  and fixed  $N$  both fail as  $d \rightarrow \infty$  ( $k \geq 2$ ). In E2 ( $k = 2, d = 64, \eta_0 = r = 1$ )  $d^2 = 4096$  against  $BN^2 \geq 16384$ ; the E5 cell  $\eta_{\Gamma} = 10\eta, N = 16$  violates it ( $BN^2/r = 1638 < d^2$ , constants aside), consistent with SGD being 24% slower than the flow there (2.26M vs 1.83M steps).

**Conjecture 1** (SGD transfer). (a) Under the conditions of Theorem B, with  $\eta = \eta_0 d^{-2k}$ ,  $\eta_{\Gamma} = r\eta$  and  $\eta_{\Gamma} \ll BN^2 d^{-2k}$  (i.e.  $B \gg r\eta_0/N^2$ ),  $\eta \cdot T_{0.5}/\tau_{\text{esc}} \rightarrow 1$  in probability. (b) In the setting of Theorem A(i) with  $m_0 \asymp d^{-1/2} < m^*$ , for every  $B$  and every  $\eta \leq \eta_0 d^{-k}$ ,  $\Pr(\max_{n \leq e^c d m^{*2}} m_n \geq m^*/2) \rightarrow 0$ ; this would follow from a supermartingale bound for  $e^{a d m^2}$  on  $\{m \leq m^*\}$  (negative population drift plus spherical drag), in the manner of Ben Arous et al. (2021).

## C. Experimental details

All toy experiments: float64 numpy, analytic gradients verified by finite differences (relative error  $< 10^{-4}$ ), online spherical SGD with fresh data each step,  $B = 32$  prompts per step unless stated,  $\eta = \eta_0/d^2$ . Projected samplers (exact in distribution) were used where the gradient depends on  $x$  only through a few projections. Compute: Exp. 1 6.4, Exp. 1b 2.8, Exp. 2 3.2, Exp. 3 2.3, Exp. 4 6.0, Exp. 5 6.0 CPU-hours on 4 cores.

**Exp. 1 (single skill, uniform initialisation).**  $k^* \in \{1, 2\}$ ,  $d \in \{8, \dots, 128\}$ , 5 seeds,  $N = 128$ , fixed  $\gamma = 0.1$  (A) and  $a = 1$  (B). Slopes of  $\log T_{0.5}$  vs  $\log d$ : B  $k^* = 1$ :  $1.98 \pm 0.06$  ( $d \geq 16$ ; the pre-registered 1.5 was an error: for  $\kappa < 2$  the drift is  $O(1)$  and  $T \propto 1/\eta = d^2$ ); B  $k^* = 2$ :  $2.33 \pm 0.22$ ; A  $k^* = 1$ :  $2.71 \pm 0.43$ ; A  $k^* = 2$ : 3.1–3.5 (20 of 105 runs censored at  $3 \cdot 10^6$  steps). Regression of  $\log(T_{0.5}\eta)$  on  $\log |m_0|$ : A  $k^* = 2$ :  $-2.19 \pm 0.08$  (reached runs with  $\eta_0 \in \{1, 2\}$ ,  $d \geq 16$ ;  $-2.29 \pm 0.07$  on all reached runs), i.e.  $\kappa \approx 4.2$ –4.3, a lower bound because of censoring; B  $k^* = 2$ :  $-0.57 \pm 0.06$  (logarithmic,  $\kappa = 2$ ); A  $k^* = 1$ :  $-0.50 \pm 0.06$ ; B  $k^* = 1$ :  $-0.13 \pm 0.02$ .

**Exp. 1b (fixed  $m_0$ ).** Threshold grid:  $\gamma \in \{1, 0.1\}$ ,  $N \in \{32, 128, 512\}$ ,  $d \in \{16, 32, 64\}$ , 3 seeds,  $3 \cdot 10^5$  steps. Escapes at  $\gamma = 1$ :  $N = 32$ : 2/3, 0/3, 0/3;  $N = 128$ : 3/3, 1/3, 0/3;  $N = 512$ : 3/3, 3/3, 3/3 (corrected  $d^* = 6.8, 30.8, 126.8$ ).  $\gamma = 0.1$ : 27/27. P7:  $\gamma = 1$ ,  $N = 128$ ,  $d = 64$ ,  $B \in \{32, 256, 1024\}$  with  $3 \cdot 10^5, 10^5, 3 \cdot 10^4$  steps: 0/9 escapes, final  $|m|$  0.005–0.015 ( $B \leq 256$ ) and 0.063–0.067 ( $B = 1024$ ). Exponent table ( $d = 32$ , 4 values of  $m_0$ , 3 seeds,  $\eta = 1/d^2$ ,  $\Gamma_0 = 0.01$  or  $\rho_0 = 0.01$ ): in Fig. 1; censored runs ( $N = 32$ : 3/12 for both free protocols;  $N = 128, 10\eta$ : 2/12) are all at the smallest  $m_0$ .

**Censored fits and absolute-time ratios.** For the three E1 cells containing censored runs (all at the smallest  $m_0$ ) we refit  $\log(T_{0.5}\eta) = a + b \log m_0$  by a Tobit likelihood with right-censoring at the step cap (‘results/README\_stats.md’):  $\kappa_{\text{eff}}$  becomes  $6.05 \pm 0.18$  (free  $\eta_{\Gamma} = \eta$ ,  $N = 32$ ; ODE 6.47),  $6.73 \pm 0.18$  (free  $10\eta$ ,  $N = 32$ ; ODE 7.02) and  $6.67 \pm 0.37$  (free  $10\eta$ ,  $N = 128$ ; ODE 6.18); the other nine cells are unchanged. All twelve are within 0.49 of the ODE and ten within 2 s.e.; bootstrap s.e. in the censored cells are 0.21–0.49. The Tobit fit assumes censored runs would eventually cross  $m = \frac{1}{2}$ , which is an assumption (their final  $|m|$  is 0.04–0.11). Absolute-time ratios median( $T_{0.5}$ )/ODE: E1 (45 reached cells, ODE at each run’s exact  $m_0$ ): 24/35/45 within 10/20/50%, range 0.64–1.34; E2 (18 cells, pinned  $\gamma = 1$  excluded as the ODE predicts a trap): 17/18/18, range 0.85–1.10. Single seeds range 0.50–2.08 $\times$ . One censored E1 cell (free  $10\eta$ ,  $N = 128$ ,  $m_0 = 0.5d^{-1/2}$ : 1/3 reached) is at least 2.35 $\times$  slower than the ODE.

**Exp. 6 (three regimes).**  $d = 64$ ,  $m_0 = d^{-1/2}$ ,  $\eta = 1/d^2$ ,  $N \in \{16, 64, 256\}$ , schemes  $NB = 4096$  and  $B = 64$ , protocols pinned  $\gamma \in \{1, 0.1\}$ , free ( $\Gamma_0 = 0.01$ ,  $\eta_{\Gamma} = \eta$ ), tied ( $\rho_0 = 0.01$ ), 3 seeds,  $10^6$  steps ( $2.5 \cdot 10^6$  for the free  $N = 16$  rerun). ODE predictions were computed and committed before the run. 12 of 72 runs are duplicates ( $N = 64$  has  $B = 64$  in both schemes). Compute 2.4 CPU-h. Medians of three seeds resolve escape times to roughly 5%. All ODE values in this appendix are LSODA (‘scripts/ode\_lsoda.py’); the SGD/ODE ratio statistics above were computed with the earlier Euler values and would rise by 0.6–3.6%.

**Exp. 7 (controls, E5).** As Exp. 6 with  $B = 64$ , 3 seeds, cap  $2.5 \cdot 10^6$  ( $10^6$  for the tied  $\rho_0 = 0.3$  cell): free  $\Gamma_0 = 0.01$ ,  $\eta_{\Gamma} = 10\eta$ : 2.26M/498k/212k (ODE 1.81M/500k/205k); free  $\Gamma_0 = 0.1$ ,  $\eta_{\Gamma} = \eta$ : 1.52M/271k/166k (ODE 1.49M/270k/161k); tied  $\rho_0 = 0.1$ : 37k/18k/17k (ODE 33k/19k/17k; one  $N = 16$  seed at 567k); tied  $\rho_0 = 0.3$ :

716k/27k/21k (ODE: trap/31k/21k); pinned  $\gamma = 0.01$ : 1.71M/1.64M/1.62M (ODE 1.70M/1.64M/1.63M);  $\eta \propto B$  at pinned  $\gamma = 0.1$ ,  $N = 64$ : 773k/192k/48k steps for  $B = 16/64/256$ , flow time 47.2/47.0/46.9. Compute 2.2 CPU-h.

**Exp. 2 (many skills).**  $P = 16$ ,  $\pi_p \propto p^{-1.5}$ ,  $M = 64$ ,  $\Gamma = 0.1$ ,  $N = 128$ , 3 seeds (plus 2 extra and  $\alpha = 1$ ). Skills 1–8 learned in every run; emergence of skills 9–16 partial. In-weight baselines: (i) readout  $\rho = \sum_p a_p/M$  (the readout-1 baseline does not learn); (ii) the 2-homogeneous student  $\sum_j \|u_j\|^2 \sigma_2(\hat{u}_j \cdot x)$ ,  $\|u_j\|_0^2 = 0.01$ ,  $\eta \in \{0.25, 1, 4\}/d^2$ , 3 seeds each: slope  $0.75 \pm 0.09$  (1000-step log),  $0.87\text{--}0.89 \pm 0.09$  (100-step log),  $T_p \eta a_p = 0.23\text{--}0.26$ ; 18.7, 10.7, 9.0, 6.3, 2.7, 3.0, 2.7, 3.7 features on skills 1–8, best single alignment  $0.86 \rightarrow 0.59$ , pooled alignment 1.000 for all 16 skills, population MSE  $3 \cdot 10^{-8}$ : collective fit. Both are collective at  $M = 64 > d$ ; a decoupled in-weight regime would need  $M < d$  (as in Ben Arous et al. (2025), whose plateau length  $\log(d/r_s)$  comes from the Gram matrix  $WW^\top$  having eigenvalues  $\approx r_s/d$  at initialisation—at  $M \geq d$  that matrix is already full rank, which is our reading of why no search plateau and no single-feature emergence appear) and was not run. Slopes and exponents as in Sec. 4; the  $\alpha = 1$  slope is  $0.88 \pm 0.09$  (A) vs  $0.61 \pm 0.08$  (B).

**Exp. 3 (composition).**  $P = 2$ ,  $\pi = (0.75, 0.25)$ ,  $M = 4$ ,  $\Gamma = 0.1$ ,  $N = 128$ , 5 pre-registered seeds + 5 extra + 3 at  $\Gamma = 0.3$ . Additivity residual with a paired estimator (same inputs, 17 queries per context):  $\max |R| \leq 0.0086$  over 13 runs (the pre-registered independent-set estimator was too noisy, 0.04–0.35). Final neuron assignments (skill 1, skill 2): (4, 0), (2, 2), (2, 2), (2, 1), (2, 2) for the pre-registered seeds. Neuron arrival times  $T \approx K/(\eta \pi_p m_0^2)$  with  $K = 0.60$  (0.54–0.78).

**Exps. 4–5 (transformers).** Tokens  $[x_i; y_i]$ , query  $[x_q; 0]$ ; embedding Linear( $d+1 \rightarrow 64$ ), 2 pre-LN blocks (4-head causal softmax attention + ReLU MLP), linear readout; Adam  $10^{-3}$ , fp32, online data, eval every 200 steps on 2048 held-out prompts,  $T_e$  = first eval with normalised MSE  $< 0.5$ . Exp. 4: MLP width 256,  $d \in \{16, 32\}$ ,  $N \cdot B = 8192$ . Exp. 5:  $d = 256$ , width 32, cells (a)  $N=256, B=64$ :  $T_e = 4000, 2800$ , third seed censored at 17k on the plateau; (c)  $N=16, B=64$ : none in 30k (3/3); (d)  $N=16, B=1024$ : 14800, 10600, 11600; (e)  $k^* = 1$ ,  $N=16, B=64$ : 1400, 1200; (f) in-weight: 400, 200; initial  $\max_j |\cos(W_j, v)|$ : 0.126, 0.152, 0.176. Exp. 9 ( $k^* = 3$ , single neuron, sweep9.py):  $\eta = 2 \cdot 10^{-4}$  for all  $d$ ,  $B = 64$ ,  $m_0 = d^{-1/2}$  exactly, cap  $10^6$  steps, 3 seeds, 27 runs, 0.89 CPU-h; medians/ODE tied  $N=128$ : 1.18, 0.89, 0.80 ( $d = 8, 16, 32$ ), pinned  $\gamma=0.2$   $d=8$ : 1.02; post-hoc controls:  $d=8$  at  $\eta/4$ : 1.007;  $d=32$  at  $B=256$ : 1.14 (declared before running). Exp. 8 (transformer\_exp8.py, same model/data code as Exp. 5): readout frozen at its initial norm (0.56–0.58) or at  $0.1 \times$  it, or trainable; cells  $N=16, B=64$ ;  $N=16, B=1024$ ;  $N=256, B=64$ ; 2 seeds; 30k steps; 5.5 CPU-h; one pinned-small  $N=256$  seed censored at 17k (budget).

## D. Pre-registration summary

All predictions were written to a version-controlled document before the corresponding run; the full text with timestamps, outcomes and changelog is in the repository (docs/preregistration.md). Summary:

Context length and readout parameterisation select the emergence exponent of in-context skills

| ID     | prediction                                                                                                                                                                 | outcome                                                                                                                                                                                                                                                                 |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1     | MC drift of A, $k^*=2$ , $\propto m^3$ ; closed form within 20%                                                                                                            | met (slope $3.14 \pm 0.07$ ; $\leq 3\%$ at $m \geq 0.2$ )                                                                                                                                                                                                               |
| P2     | $\gamma=1$ , $N=128$ : stuck for $d \geq 32$                                                                                                                               | partly ( $d=64$ stuck, $d=32$ : 1/3 escaped); threshold formula later corrected, $d^* \approx 31$                                                                                                                                                                       |
| P3     | $\gamma=0.1$ : escapes at all $d$                                                                                                                                          | met (27/27)                                                                                                                                                                                                                                                             |
| P4     | free readout: $\kappa$ rises as $N$ falls, $4 \rightarrow 8$                                                                                                               | direction met; magnitudes match the ODE at $d=32$ ( $4.85 \rightarrow 7.0$ ), not the asymptotes                                                                                                                                                                        |
| P4'    | tied readout: $\kappa \approx 4$ at all $N$ ; $\rho_0=1$ stuck for $d \geq 32$ (flow traps; SGD delays)                                                                    | $\kappa$ met (4.10–4.44); $\rho_0=1$ : $d=32$ escaped 3/3, $d=64$ stuck                                                                                                                                                                                                 |
| P5     | learned $\Gamma \approx \Gamma^*(0.5; N)$                                                                                                                                  | not met (relaxation too slow at $\eta_\Gamma = \eta$ )                                                                                                                                                                                                                  |
| P6     | A vs B slope difference $\approx 1$ at $k^*=2$                                                                                                                             | met via $m_0$ -regression ( $\kappa$ 4.2 vs 2); $d$ -slopes $0.9 \pm 0.5$                                                                                                                                                                                               |
| P7     | no escape at any $B$                                                                                                                                                       | met (0/9)                                                                                                                                                                                                                                                               |
| P8     | transformer, $d=32$ : $T_e$ falls $\geq 3\times$ from $N=16$ to 256 at fixed tokens                                                                                        | <b>failed, reversed</b> (1000 $\rightarrow$ 2800)                                                                                                                                                                                                                       |
| P9     | $k^*=1$ weak $N$ -dependence; in-weight fastest                                                                                                                            | partly ( $\times 2.0$ ; in-weight fastest)                                                                                                                                                                                                                              |
| P10    | $N$ -dependence in both softmax and linear attention                                                                                                                       | met (same direction)                                                                                                                                                                                                                                                    |
| P11a–b | $d=256$ , width 32: (c) stuck, (d) stuck                                                                                                                                   | (c) stuck 3/3; (d) emerged 3/3 at $\approx 3\times$ (a): <b>both rival and default failed</b>                                                                                                                                                                           |
| P11c   | $k^*=1$ and in-weight controls fast                                                                                                                                        | met                                                                                                                                                                                                                                                                     |
| P12    | additivity residual $ R  \leq 0.02$                                                                                                                                        | failed as pre-registered (estimator noise); identity confirmed with a paired estimator                                                                                                                                                                                  |
| P13    | no extra compositional delay                                                                                                                                               | supported with substitute statistic (floor made the original unreachable)                                                                                                                                                                                               |
| P14    | multiplicative accuracy, Gaussian closed form                                                                                                                              | approximately multiplicative; Gaussian form wrong                                                                                                                                                                                                                       |
| P15    | product composition not learned                                                                                                                                            | met                                                                                                                                                                                                                                                                     |
| P19    | $d=64$ : pinned-1 trapped; free $T \propto \text{ODE}$ with $T(16)/T(256) = 4.4$ ; tied flat within 5%                                                                     | pinned-1: 0/9 at $N \leq 64$ , 3/6 at $N=256$ ; free within one s.e. of ODE, ratio 4.4; tied flat within $\times 1.25$ (5% not resolvable)                                                                                                                              |
| P20    | controls: ratios 8.9/9.1 (free $10\eta/\Gamma_0=0.1$ ), tied $\rho_0=0.1$ ratio 1.9, tied $\rho_0=0.3$ traps at $N=16$ , pinned 0.01 flat, $\eta \propto B$ flow-invariant | 10.7/9.2; 2.15; <b>escaped after <math>30\times</math> delay (failed as worded)</b> ; 1.06; invariant to 0.5%                                                                                                                                                           |
| P21    | transformer readout protocols: pinned $O(1) \Rightarrow N=16$ stuck at $B=1024$ ; pinned small $\Rightarrow N=16$ , $B=64$ emerges; free reproduces Exp. 5                 | <b>failed</b> (8400, 5800 at $B=1024$ ); <b>failed</b> (stuck 2/2); held ( $3.7\times$ , $3.8\times$ )—kill criterion met                                                                                                                                               |
| P22    | $k^*=3$ : tied $\rho_0=0.01$ secants 6.5/6.4, medians within 15%; tied $N$ -ratio 1.16/1.63/ $\infty$ ( $d=8/16/32$ ); pinned $\gamma=0.2$ : $d=8$ escapes, 16, 32 trapped | secants 5.7/6.0 (kill not triggered); <b>medians</b> 1.18/0.89/0.80 ( <b>failed; controls: 1.007 at <math>\eta/4</math>, 1.14 at <math>B=256</math></b> ); 1.14/19/ $\infty$ ( <b>failed at <math>d=16</math>: 47k/500k/censored</b> ); held (1.02; 3/3 censored twice) |
| P16    | $T_p$ slopes 1.5 (A), 0.75 (B)                                                                                                                                             | met (1.44–1.53; B 0.75 capacity-matched, 0.75–0.89 2-homogeneous)                                                                                                                                                                                                       |
| P17    | A steps $\geq 2\times$ sharper                                                                                                                                             | met ( $3.8\times$ )                                                                                                                                                                                                                                                     |
| P18    | loss exponents 0.33 (A), 0.67 (B)                                                                                                                                          | A within tolerance (0.40); B failed (collective fit saturates)                                                                                                                                                                                                          |